

# COVID-19, Migrants, and Automation in China <sup>\*</sup>

Jingcheng Jiang <sup>†</sup>

Yalan Li <sup>‡</sup>

## Abstract

We study whether transient policy-induced labor supply shocks can permanently redirect technological change. China's zero-COVID mobility restrictions provide a natural experiment: between 2020 and 2022, hundreds of millions of migrants were blocked from urban manufacturing centers, generating a nearly 60% collapse in migrant labor supply. Exploiting cross-county variation in pre-pandemic migrant dependence, we find that more exposed counties experienced sharp post-COVID growth in robotics patenting, with no corresponding change in total patenting activity. We also construct a Bartik IV and find consistent results: uncertainty in origin regions disrupted labor flows to destinations, and counties losing more migrants saw larger increases in automation patents. The response is concentrated in tight local labor markets where native workers could not substitute for absent migrants, consistent with binding and recurrent quantity constraints rather than wage-driven substitution. Displaced migrants bear persistent earnings and employment losses. The results show that temporary restrictions on labor mobility can have lasting effects on the direction of innovation.

**JEL Codes:** J61, O33, R23, J24, O14

**Keywords:** Migration, Labor Shortage, Automation, COVID-19, Directed Technological Change

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<sup>†</sup>Graduate School of Economics, University of Tokyo. Email: jingcheng.jiang@uqconnect.edu.au

<sup>‡</sup>Faculty of Economics, Kanto Gakuen University. Email: morethanlan@gmail.com

# 1 Introduction

While directed technical change theory predicts that factor scarcity redirects innovation toward labor-saving technologies (Acemoglu, 2002), the empirical literature has largely focused on settings where labor supply adjusts gradually, through immigration waves (Lewis, 2011; Danzer et al., 2024) or demographic transitions (Acemoglu et al., 2022). This paper asks whether sudden labor supply shocks, compounded by persistent policy uncertainty over workers’ mobility, can similarly redirect innovation. We argue that when unexpected mobility restrictions can instantly sever firms’ access to workers, human capital ceases to be a flexible input and becomes a source of acute operational fragility. Under such conditions, automation serves not merely as a factor substitution response but as a hedge against operational risk. Using China’s COVID-19 mobility restrictions as a sudden and exogenous shock to internal migration flows, we identify how unpredictable labor supply constraints drive the country’s rapid and persistent surge in industrial automation.

China provides an ideal setting for isolating this high-frequency mechanism for three reasons. First, it operates the world’s largest system of internal labor migration: approximately 292 million rural *hukou* holders, representing 39% of the national workforce, form the backbone of China’s manufacturing sector.<sup>1</sup> Second, the enforcement of China’s zero-COVID policies replaced a stable labor market with extreme policy uncertainty. Because local governments frequently initiated abrupt, zero-warning lockdowns to meet strict political mandates, human labor became fundamentally unpredictable. This triggered an unprecedented collapse in the out-migration rate of more than half relative to its pre-pandemic level.<sup>2</sup> Third, China dominates global automation: robot installations nearly doubled between 2019 and 2022 while the rest of the world stagnated (Figure 1). Whether this surge was caused by the labor supply collapse, or merely coincided with it, is the central question we address.

To analyze these dynamics, we construct a dataset that spans two levels of aggregation, linking individual labor supply decisions to regional innovation outcomes. At the individual level, we use the China Family Panel Studies (CFPS) to track migration decisions and employment status across the 2018, 2020, and 2022 waves, thereby identifying the specific demographic groups most affected by mobility restrictions. At the county level, we construct a panel of patent applications from the China National Intellectual Property Administration (CNIPA) covering more than 40 million records spanning 2,200 counties

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<sup>1</sup>Data from the National Bureau of Statistics of China (National Bureau of Statistics of China, 2021). The *hukou* system creates institutional segmentation that makes migrant mobility observable, regulable, and sharply affected by policy.

<sup>2</sup>The 76-day Wuhan lockdown (January 2020) and the 61-day Shanghai lockdown (spring 2022) imposed sudden, severe constraints on worker movement, reinforced by a color-coded QR code system embedded in WeChat and Alipay that required a green code to access workplaces and public transport.

over a ten-year period. By classifying patents using text-based algorithms targeting keywords such as “robot” and “manipulator” (robotic arm), this dataset allows us to measure the direction of indigenous technical change at a granular spatial level and test whether the pandemic redirected innovation toward labor-saving technologies rather than spurring broad inventive activity.

First, we establish the structural bias of the labor supply shock. Using individual-level survey data, we document that the pandemic reduced cross-regional mobility for workers in migrant-intensive sectors, relative to those in sectors with low migration dependence, with migrant labor supply falling by nearly 60% during the peak restriction period, as shown in Figure 2.<sup>3</sup> Critically, this contraction was not a generalized withdrawal of labor. The shock fell exclusively on low-skilled workers, effectively severing the supply of routine manual labor to urban manufacturing centers while high-skilled professionals remained largely insulated. Firm-level evidence from the Enterprise Survey on Innovation and Entrepreneurship in China (ESIEC), collected in early 2020, corroborates this picture: comparing firms within the same city and industry, those with a higher share of migrant workers in their pre-pandemic workforce were significantly less likely to have resumed operations, faced more severe labor shortages, and attempted to raise wages to attract or retain workers.<sup>4</sup>

Second, we provide causal evidence that the labor supply shock redirected innovation toward automation. Counties more reliant on migrant labor experienced significant post-COVID increases in robotics and automation patenting, while non-automation patents remained unchanged, confirming directed technical change rather than a generalized innovation boom (Acemoglu, 2002). This pattern is corroborated by cross-country robot installation data: Chinese robot installations nearly doubled between 2019 and 2022 while the rest of the world stagnated, with the surge concentrated in migrant-intensive sector groups such as automotive and manufacturing and construction, rather than in the already-robotized high-tech industries. Firm-level evidence further shows that firms in migrant-dependent regions increased both R&D expenditure and fixed capital per employee after the shock. Heterogeneity analysis reveals that the innovation response is concentrated in counties with already-tight labor markets, where firms could not substitute missing migrants with local workers (Acemoglu and Restrepo, 2018; Danzer et al., 2024) — precisely the conditions under which labor scarcity theory predicts automation investment

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<sup>3</sup>Manufacturing employment is concentrated in coastal and urban industrial counties, far from the rural interior where most workers originate. Agricultural workers, by contrast, tend to work near their place of origin and are therefore less reliant on long-distance mobility.

<sup>4</sup>These patterns confirm that COVID-19 mobility restrictions created a binding labor supply constraint for the firms that actually employed migrant workers, rather than a broader demand-side shock common to all firms in affected regions, supporting the identification strategy in the main analysis.

to be most compelling (Acemoglu, 2010). Importantly, we systematically rule out four alternative explanations: social distancing imperatives, export surges, local government industrial policy, and demand-side shocks; each fails to account for the observed automation response.

Finally, we trace the distributional consequences of the pandemic-induced automation episode (Appendix A.4). Pre-pandemic migrant workers experienced persistent increases in informal employment and income declines of 8 to 9 percent relative to non-migrants, bearing most of the adjustment costs (Dix-Carneiro and Kovak, 2019; Ponczek and Ulyssea, 2022). Importantly, these losses persist well into 2022, after mobility restrictions were lifted, suggesting that the costs reflect not merely the direct disruption to spatial matching but a more structural labor market deterioration consistent with automation displacement. While we cannot fully decompose the two channels, the persistence of effects beyond the restriction period points to automation as a meaningful contributor. Transient mobility frictions, our findings suggest, can generate not only permanent industrial upgrading but also persistent distributional costs concentrated among the most vulnerable workers.

This paper contributes to four strands of literature. First, we provide direct causal evidence for theories of directed technical change and identify the conditions under which sudden factor supply shocks redirect innovation. The idea that factor scarcity shapes the direction of innovation dates to Hicks (1932) and Habakkuk (1962) and was formalized by Acemoglu (2002), in which a price effect and a market size effect jointly govern the direction of technical change.<sup>5</sup> Empirically, gradual factor supply changes predictably shift innovation toward labor-saving technologies (Danzer et al., 2024; Hémous et al., 2025), and exogenous shocks can redirect firm-level patenting more broadly (Azoulay et al., 2019). For sudden shocks, however, the evidence is mixed: the abrupt Bracero exclusion did not trigger technology adoption (Clemens et al., 2018), indicating that a sudden quantity constraint alone is not sufficient. Our contribution is to show when it is. In our setting, automation responded where three conditions held jointly: the constraint was binding, in tight labor markets with no local substitutes; the disruption was recurring and unpredictable rather than one-off,<sup>6</sup> making automation a hedge against future stoppages as well as a substitute for absent workers; and automation technology was feasible and financially accessible (Acemoglu, 2010).

Second, we contribute to the literature on migration, labor supply, and firm adjustment. Lewis (2011) shows that abundant low-skilled labor suppresses automation in U.S.

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<sup>5</sup>See also Kennedy (1964), Acemoglu (2010), and Acemoglu and Restrepo (2018) for theoretical extensions.

<sup>6</sup>When a shock is one-off and expected to be temporary, the option value of waiting dominates: firms can bridge the disruption with inventories, overtime, or delayed orders and re-hire workers once it passes, rather than commit to irreversible investment (Bloom, 2009). It is the prospect of repeated, unannounced stoppages that eliminates the value of waiting and makes automation the risk-reducing choice.

manufacturing, and [Dustmann and Glitz \(2015\)](#) demonstrate that labor supply shocks trigger systematic reorganization of production across the skill distribution. The two most closely related papers are [Imbert et al. \(2022\)](#) and [Danzer et al. \(2024\)](#). [Imbert et al. \(2022\)](#) find that rural-urban migrant inflows into Chinese manufacturing cities make production more labor-intensive. [Danzer et al. \(2024\)](#) exploit a German immigrant allocation policy to show that exogenous increases in low-skilled labor supply reduce automation patenting. Both papers study positive labor supply shocks that suppress automation; we study the causal inverse, identifying how a sudden and binding labor supply collapse triggers it. [Peri \(2012\)](#) provides complementary evidence that exogenous labor supply shifts redirect task specialization among native workers. Studies of internal migration in China further document large positive wage externalities on urban natives ([Combes et al., 2015](#)) and substantial mobility frictions that suppress aggregate productivity ([Tombe and Zhu, 2019](#)). What distinguishes our setting is that China’s dependence on long-distance rural-urban migration, structured by the *hukou* system, created a labor market uniquely vulnerable to mobility restrictions, transforming a public health shock into a binding and geographically concentrated labor supply collapse with no counterpart in the existing migration-and-firms literature.

Third, we speak to the literature on economic crises and structural change. [Jaimovich and Siu \(2020\)](#) showed that jobless recoveries since 1990 stem from the permanent elimination of routine occupations during recessions, as downturns accelerate the displacement of middle-skill workers by technology. [Hershbein and Kahn \(2018\)](#) documented that skill requirements in job postings rose sharply in metropolitan areas hit hardest by the Great Recession, consistent with crisis-induced restructuring.<sup>7</sup> We extend this literature by demonstrating that public health shocks, operating through labor supply rather than product demand, can similarly trigger durable technological upgrading. The pandemic setting is distinctive in this respect: unlike demand-driven recessions, COVID-19 lockdowns constrained labor availability while leaving capital costs and product demand relatively intact, isolating the supply-side mechanism and allowing us to attribute the automation response to factor scarcity rather than demand compression.

Finally, we contribute to a growing literature on robot adoption in China. Previous studies have documented the causes of China’s rapid robotization, emphasizing rising labor costs and demographic pressures ([Cheng et al., 2019](#); [Fan et al., 2021](#)), as well as its consequences for firms’ skill composition ([Tang et al., 2021](#)), worker outcomes ([Giuntella et al., 2025](#)), and workplace safety ([Luo et al., 2025](#)). We complement this literature by identifying a distinct mechanism: China’s zero-COVID policies imposed sudden and severe

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<sup>7</sup>Related work includes [Charles et al. \(2016\)](#) on manufacturing decline and [Autor and Dorn \(2013\)](#) on job polarization.

cross-regional mobility restrictions that prevented migrant workers from reaching urban manufacturing centers, generating a binding labor supply collapse that the prevailing neoclassical accounts cannot explain. The *hukou* system is central to this story not as a direct constraint, but because it created a labor market structure uniquely dependent on long-distance mobility, amplifying the disruptive impact of travel restrictions. Our findings suggest that firms automated not simply because labor became expensive, but because it became unavailable, reframing China’s recent industrial upgrading as a response to a public health shock interacting with deep institutional features of its labor market.

The paper is organized as follows: Section 2 describes detailed background information on COVID-19 and internal migrants in China; Section 3 and Section 4 describe the data and identification strategy; Section 5 presents the baseline results; Section 6 presents additional results that further shed light on the mechanism; Section 7 concludes.

## 2 Background

### 2.1 COVID-19 Lockdowns and Mobility Controls

The COVID-19 pandemic led to a series of unprecedented public-health interventions in China, including large-scale lockdowns and mobility restrictions that sharply disrupted labor flows. The first and most extensive intervention was the lockdown of Hubei Province, enacted on January 23, 2020, and lasting until April 8, 2020, a total of 76 days. During this period, outbound transportation was suspended and residents were required to remain at home, halting economic activity in one of China’s major industrial and transportation hubs.

Under the “zero-COVID” strategy that governed policy until December 2022, subsequent outbreaks triggered repeated restrictions of this kind, most prominently the phased Shanghai lockdown of spring 2022, which culminated in a full citywide closure lasting roughly 61 days. These headline episodes, however, were not isolated events. Using a lockdown-event dataset we assemble for all 366 prefecture-level cities (Appendix Table C.4), we document that the average city experienced 4.8 lockdown events between 2020 and 2022 and that over a quarter of all cities faced at least one full citywide lockdown. Nor did the restrictions fade after the initial outbreak: 2022, the third year of the policy, was the most severe, accounting for more than half of all recorded events. Lockdowns were therefore not a temporary disruption concentrated in early 2020 but a recurring feature of the economic environment for three full years, and neither firms nor workers could predict when or where the next restriction would arrive.

Between formal lockdowns, mobility remained regulated by a nationwide digital



enforcement infrastructure. As shown in Figures B.2 and B.3, a three-color “health code” system embedded in the ubiquitous WeChat and Alipay applications assigned each citizen a QR code based on travel history and exposure risk. A green code was required to access workplaces, public transportation, and commercial venues; maintaining it required frequent nucleic-acid testing, often every 48 to 72 hours, while a yellow or red code imposed mandatory quarantine of 7 to 14 days or confinement to one’s residential compound. The detection of positive cases could trigger lockdowns at the neighborhood or compound level without warning. For rural migrants attempting to return to manufacturing hubs after the Chinese New Year, this system served as a near-complete barrier to the resumption of labor supply. The administrative intensity of the regime is visible in the language of local government policy documents: keywords related to epidemic prevention and quarantine surged after 2020 and remained elevated through 2023 (Figure B.5).

## 2.2 Migrant Workers in China

China’s development over the past four decades has been closely intertwined with the large-scale movement of low-skilled rural workers into urban labor markets. Migrant workers, defined as individuals with rural household registration (*hukou*) who engage in non-agricultural work outside their home township for at least six months, form the backbone of China’s manufacturing, construction, and urban service industries: in 2021 they numbered 292.51 million, approximately 39.2% of the national workforce (National Bureau of Statistics of China, 2021), and roughly 60% work away from their home county, many in the coastal manufacturing centers of Guangdong, Zhejiang, and Jiangsu.

The *hukou* system shaped the form this migration took. Because access to public services, schooling, and housing remains tied to the place of registration, most migrants cannot settle permanently at their destination. They work as long-term sojourners instead, living in dormitories or rented rooms and returning home at least once a year around the Chinese New Year, in the world’s largest annual human migration. This vast pool of mobile, low-skilled labor underpinned China’s comparative advantage in labor-intensive production.

The same circular structure made migrant labor uniquely vulnerable to mobility restrictions. The first outbreak struck in late January 2020, precisely during the New Year travel period, when hundreds of millions of migrants had already returned to their home villages; lockdowns and travel controls then prevented them from returning to their jobs, and in the following years health codes and recurring local lockdowns kept the cost and uncertainty of long-distance travel high. As we document in Section 5, the share of workers employed away from their hometown collapsed by more than half after 2020 and had not recovered

by 2022 (Figure 2).

## 2.3 Labor Scarcity and Uncertainty for Firms

Migrant workers were the group most exposed to the mobility regime described above, for reasons rooted in their labor market position. They are typically hired under short-term or informal contracts, receive no wages during production shutdowns, and have weak access to urban social protection systems, while facing high housing and living costs in destination cities (He et al., 2022). Confronted with uncertain employment prospects and the risk of repeated lockdowns, many chose to return to their hometowns rather than remain in cities without stable income; survey evidence documents a renewed “returning-home tide” during the pandemic (Ma et al., 2024), and mobile-phone data show that migrants’ return to work after outbreaks was slow and geographically uneven (Liu et al., 2023).

For firms, the consequence was not only fewer workers but an unpredictable labor supply. Even outside formal lockdowns, a positive test among employees could suspend production, and infection clusters could shut down entire industrial parks or neighborhoods. Firms reliant on on-site labor therefore faced persistent difficulties coordinating shifts, maintaining stable production lines, and ensuring the continuous availability of workers.

Together, these conditions define the shock we study: a sharp contraction in migrant labor supply combined with three years of uncertainty over whether workers already employed would be able to reach the factory. In sectors heavily dependent on low-skilled migrant workers, this environment created strong incentives to reduce reliance on physically mobile labor and to adjust production along the capital–labor margin, the response we test in the remainder of the paper.

## 3 Data

### 3.1 Migrant Labor Data

#### 3.1.1 County-level Migration Exposure

To capture both the regional exposure to labor shortages and the individual-level labor supply dynamics, we construct our migrant variables using two distinct data sources. First, to measure the spatial variation in pre-pandemic reliance on migrant labor, we construct a county-level measure of migrant exposure using the 2015 China 1% Population Survey (mini-census), following Heblich et al. (2022); Guo et al. (2024). By observing both the



registered *hukou* location (origin) and the current place of residence (destination) for each surveyed individual, we aggregate these records and scale them by survey sampling weights to recover population-level bilateral migration flows. This procedure allows us to calculate the migrant share of each county’s total population in 2015, which serves as our primary treatment variable capturing local structural dependence on mobile labor prior to the COVID-19 shock. Since this measure is constructed from pre-pandemic data, it is predetermined with respect to the COVID-19 shock, making it suitable as a measure of structural exposure rather than an endogenous response to the pandemic. We report the most migrant-intensive counties based on this distribution in Table C.5.

### 3.1.2 Individual-level Migration Decision during COVID-19

Second, to independently identify the impact of the COVID-19 pandemic on individual labor supply decisions, we employ data from the China Family Panel Studies (CFPS). Administered by Peking University, the CFPS is a nationally representative longitudinal survey has been conducted biennially since 2010. Its baseline sample covers 25 provinces, municipalities, and autonomous regions, representing approximately 95% of China’s population. The 2010 baseline survey interviewed 14,960 households and 42,590 individuals, who have since been followed over time. As in most panel surveys, some respondents attrit in each wave, while refresher samples are also added. By the end of the 2018 survey, the CFPS had completed interviews with about 15,000 households and collected roughly 44,000 individual questionnaires. The survey contains rich information on demographics, education, employment, income, health, migration history, and family relations.

Utilizing the 2018, 2020, and 2022 waves, we define migrant status based on the survey question: “In the past 12 months, have you worked away from your hometown?”. The data reveal a structural collapse in mobility: the prevalence of migrant workers declines sharply across waves, from 26.7% in 2018 to 12.2% in 2020 and 13.8% in 2022, representing declines of 54.3% and 48.3% relative to the 2018 baseline, as shown in Figure 2.

Table C.3 presents summary statistics for the full CFPS sample, stratified by pre-pandemic employment sector. On average, 18.1% of the full sample reported out-of-hometown work, with the share substantially higher among non-agricultural workers (25.1%) than agricultural workers (11.4%). The non-agricultural group is systematically more likely to be migrants: they are younger, more educated, more likely to hold non-agricultural *hukou*, and have a lower female share.<sup>8</sup> The final sample comprises approximately 18,200 individuals in the treatment group and 19,100 in the control group.

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<sup>8</sup>Specifically, the non-agricultural group is younger (41.1 vs. 51.9 years), more educated (3.51 vs. 2.29 on an eight-level scale), more likely to hold non-agricultural *hukou* (35.3% vs. 9.7%), and has a lower female share (38.2% vs. 49.2%).

## 3.2 Automation Patents

We obtain patent data from the China National Intellectual Property Administration (CNIPA), covering all patent applications from 2015 to 2024, as a measure of automation innovation following an established literature that uses patent applications to capture the direction of technological change (Danzer et al., 2024; Hémous et al., 2025).<sup>9</sup> To identify automation patents, we apply a keyword-based classification targeting robotics, manipulation-related terms, and automated guided vehicles (AGVs) in both English and Chinese, supplemented by IPC code B25J (manipulators and chambers with manipulation devices). Table C.1 and Table C.2 show the examples of keywords in the patent applications. We extract county-level assignee locations to construct annual county-level counts of automation patent applications, covering 2,200 counties over the 10-year sample period.

Table 1 presents summary statistics by migration status and period, where high-migration counties are defined as those with above-median migrant shares in 2015. Three patterns emerge. First, automation patenting increased substantially after COVID-19 in both groups: mean log robot patent counts rose from 1.105 to 1.409 in high-migration counties and from 0.538 to 0.781 in low-migration counties. Second, and most importantly, the gap between high- and low-migration counties widened post-pandemic, with the difference in log robot counts growing from 0.567 to 0.628. Critically, this divergence is specific to automation patents: the gap in log non-automation patent counts remained virtually unchanged (0.530 pre-COVID versus 0.526 post-COVID), suggesting that COVID-19 differentially accelerated automation-directed innovation in migrant-dependent regions rather than spurring broad inventive activity. Figure 9 reinforces this finding, showing a post-COVID decline in the share of counties with zero automation patents, indicating that the pandemic drew new entrants into automation patenting rather than merely intensifying effort among existing innovators. Figure B.4 further maps this geographic expansion, showing that the post-COVID surge in robot patent activity spread across a substantially larger set of Chinese counties, with new entrants concentrated in manufacturing provinces historically dependent on migrant labor. Finally, the assignee records identify who files these patents. As shown in Figure 3, robot patents are filed predominantly by firms, whose share rises from roughly 60% of filings before the pandemic to 70% after, while universities' share declines and hospitals file essentially none; among firm filers, robotics and automation specialists, machinery producers, and electronics firms dominate, with medical and biotech firms accounting for under three percent.

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<sup>9</sup>The CNIPA database contains over 40 million records with detailed information on application and grant dates, assignee identity, inventor location, IPC classifications, and citations. All design patents are excluded.

## 4 Empirical Strategy

This paper examines whether COVID-19-induced labor supply disruptions accelerated automation in China. Our primary analysis exploits within-China variation in industrial structure to identify the effect on automation patenting. We complement this with individual-level migration analysis to document the labor supply shock.

### 4.1 Main Specification: COVID-19 and Automation Patents

Our identification strategy exploits cross-county variation in pre-pandemic migrant reliance as a measure of differential exposure to COVID-19-induced labor supply disruptions. We estimate the following specification:

$$\text{AutoPatent}_{pt} = \alpha + \beta(\text{Migrant Share}_p \times \text{PostCOVID-19}_t) + \gamma_p + \delta_t + \mathbf{X}'_{p,2015} \times \delta_t + \varepsilon_{pt}, \quad (1)$$

where  $\text{AutoPatent}_{pt}$  measures automation innovation in county  $p$  at time  $t$ , using both patent shares (direction of innovation) and counts (volume).  $\text{Migrant Share}_p$  is the share of migrants in county  $p$ 's population in 2015, capturing pre-pandemic reliance on migrant labor. Figure 4 presents the spatial distribution of treatment intensity.  $\text{PostCOVID-19}_t$  is an indicator for the post-2020 period. County fixed effects ( $\gamma_p$ ) absorb time-invariant local characteristics, and year fixed effects ( $\delta_t$ ) capture aggregate shocks common to all counties. We further include interactions  $\mathbf{X}'_{p,2015} \times \delta_t$ , where  $\mathbf{X}_{p,2015}$  comprises initial GDP per capita, to control for differential trends in wealthy areas. Standard errors are clustered at the county level.

The coefficient of interest,  $\beta$ , captures the differential post-COVID change in automation patenting for counties more exposed to the migrant labor supply disruption. We validate the parallel trends assumption via event study specifications, and conduct placebo tests using non-automation patent counts to confirm that results reflect directed technical change rather than a broad increase in inventive activity.

### 4.2 Labor Supply Shock

To establish that COVID-19 disrupted labor mobility, we estimate a difference-in-differences model using CFPS panel data:

$$Y_{it} = \alpha + \beta(\text{NonAgri}_{i,2018} \times \text{PostCOVID-19}_t) + \gamma_i + \delta_t + \phi_{pt} + \varepsilon_{ipt}, \quad (2)$$

where  $Y_{it}$  indicates individual's out-of-hometown work decision over year,  $\text{NonAgri}_{i,2018}$  is an indicator equal to one if individual  $i$  was employed in non-agricultural work in

2018 and zero otherwise. We define the treatment group based on pre-pandemic non-agricultural employment status rather than baseline migrant status, for two reasons. First, non-agricultural workers are structurally more dependent on cross-regional migration than their agricultural counterparts (Krugman, 1991; Au and Henderson, 2006), making pre-pandemic sector of employment a natural intent-to-treat variable. We do not treat  $\text{NonAgri}_{i,2018}$  as an exogenous occupational assignment, since pre-pandemic sector choice may itself reflect unobserved individual characteristics correlated with migration propensity. However, this endogeneity is not central to our design, because we do not interpret the interaction term as the causal effect of non-agricultural employment on out-of-hometown work. Instead, we use pre-pandemic non-agricultural status as an intent-to-treat proxy for differential exposure to mobility restrictions, capturing ex ante differences in dependence on cross-regional migration. Second, using baseline migrant status directly as the treatment variable mechanically yields negative post-treatment coefficients, as individuals who were migrants before the pandemic are more likely to have reduced their mobility after the shock, confounding the estimated treatment effect with mean reversion. We alternatively include individual, household, year, and province-by-year fixed effects.

## 5 Results

### 5.1 COVID-19 and Labor Mobility Restriction: Manufacturing vs. Agricultural Workers

We examine whether workers in migrant-intensive sectors experienced a differential reduction in cross-regional mobility following the onset of COVID-19, relative to those in sectors with low migration dependence. The outcome variable is a binary indicator, *Out-of-Hometown Work*, equal to one if the individual worked outside their registered *hukou* location. Table 2 presents the baseline results. Across all specifications—controlling for individual, household, and province-by-year fixed effects—the coefficient for the interaction term  $\text{Non-Agricultural Worker}_{\text{pre-COVID}} \times \text{PostCOVID-19}$  is negative and statistically significant at the 1 percent level. The estimate in Column (3) indicates that relative to agricultural workers, non-agricultural workers were approximately 11 percentage points less likely to work outside their hometowns after the onset of the pandemic. This confirms that the COVID-19 shock and associated travel restrictions disproportionately disrupted the labor supply of non-agricultural workers.

A key premise of our mechanism is that the labor supply shock was concentrated among low-skilled workers, who are less able to work remotely and more vulnerable to physical mobility restrictions. Figure B.6 illustrates this pattern descriptively: the share of

migrant workers is highest among low-educated workers and declines monotonically with education level, confirming that the migrant labor force is disproportionately composed of low-skilled workers. To test this formally, we examine heterogeneity by education level in Table C.11. The results provide strong support for the low-skill hypothesis. The main interaction coefficient (representing the base group: illiterate or primary education) is large and negative. However, the triple-interaction terms with higher education levels are consistently positive and significant. For individuals with middle school education or above, the positive interaction coefficients largely offset the negative baseline effect. This implies that the reduction in cross-regional labor mobility was driven almost entirely by low-skilled workers, while high-skilled professionals remained relatively shielded from these disruptions. Additional heterogeneity analyses by *hukou* status, gender, and age are reported in Appendix A.3.

The labor supply collapse is also visible at the county level, and it loaded on the counties most dependent on migration. Table C.9 aggregates the CFPS responses to county-year cells and interacts each county’s pre-pandemic out-migration intensity, measured from the 2015 mini-census, with survey-wave indicators, absorbing county fixed effects. Counties with a 10 percentage point higher pre-pandemic out-migration share experienced a 0.4 percentage point larger decline in their out-of-hometown work rate by 2020; the 2022 interaction remains negative though less precisely estimated. Because CFPS samples households at their home location, this estimate captures the collapse of out-migration from sending counties, which transmits to migrant-receiving destination counties through pre-existing migration networks, precisely the variation exploited by the shift-share design in Appendix 5.6. This county-level first stage confirms that our treatment intensity identifies counties where labor supply actually contracted, ruling out the possibility that migrant flows into high-exposure counties were politically protected during the pandemic.

## 5.2 Correlation between Migrants and Automation before COVID-19

We verify the spatial sorting of migrants prior to the shock. Figure 5 illustrates the relationship between the migrant share and local industrial characteristics before 2020. Panel (a) shows that migrants are heavily concentrated in counties with high manufacturing value-added. Panels (b) and (c) reveal that, consistent with the directed technical change mechanism (Danzer et al., 2024), migrant-receiving counties exhibit lower pre-pandemic automation patenting—abundant cheap labor reduced firms’ incentive to invest in labor-saving technologies, which is consistent with the literature Danzer et al. (2024). This pattern confirms that the COVID-19 shock hit precisely the regions where labor scarcity was most acute and the scope for automation-driven adjustment was greatest.

### 5.3 Correlation between Migrant Dependence and Firm-level Labor Shortage During COVID-19

To further validate that migrant-dependent regions experienced more severe labor supply disruptions during the pandemic, we draw on the Enterprise Survey on Innovation and Entrepreneurship in China (ESIEC), a firm-level survey conducted in early 2020 (Barwick et al., 2025). As reported in Table 3, firms with higher migrant worker share were significantly less likely to have resumed operations and reported a significantly higher share of absent workers, consistent with acute labor shortages disrupting production.<sup>10</sup> These firms were also significantly more likely to report an inability to fulfill existing orders. The coefficient on wage increases is positive, suggesting that firms with more migrant workers were more inclined to raise wages in response to labor shortages, but the estimate is not statistically significant, so this evidence should be read as suggestive rather than conclusive. Labor cost pressure, PPE shortages, supply chain disruption, and logistics disruption are all statistically insignificant, suggesting labor costs were not a major concern for these firms and that the factor-price channel is unlikely to explain the automation response. This pattern confirms that mobility restrictions generated a labor-specific supply constraint in migrant-dependent counties, rather than a broader demand or cost shock, supporting the identification strategy in the main analysis.

### 5.4 Baseline: COVID-19 and Automation Patents

Table 4 reports the effect of COVID-19-induced labor supply disruptions on automation patenting.

#### 5.4.1 Main Findings

Across all specifications, counties with higher pre-pandemic migrant reliance experienced significantly greater post-COVID growth in automation patenting. Under the most demanding specification (Panel B), a one standard deviation increase in the migrant share (1.229 percentage points) is associated with a 0.077 percentage point increase in the robot patent share (relative to total patents), a 4.3% increase in log robot counts, a 6.7 patent increase in the raw robot patent count, and a 10.4% increase in the per capita measure.<sup>11</sup>

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<sup>10</sup>We measure each firm's migrant worker share using pre-pandemic data from 2017, prior to the onset of COVID-19, and use this as our main explanatory variable.

<sup>11</sup>Estimates are stable across Panels A and B, with coefficients ranging from 0.049–0.063 for patent share, 5.457–6.814 for raw robot patent counts, and 0.035–0.051 for log counts, suggesting results are not driven by differential trends in wealthy or historically high-migration areas.



All estimates are significant at the 1 percent level and robust to alternative functional forms, including IHS-transformed counts (columns 9–10).

These findings are consistent with the canonical directed technical change framework (Acemoglu, 2002), in which a reduction in the relative supply of a factor of production induces innovation complementary to the scarce factor’s substitute. In our setting, the sudden curtailment of migration reduced the effective labor supply in manufacturing counties, raising the price of routine labor and tilting the returns to innovation toward automation technologies. This mechanism echoes the historical evidence on skill-biased technical change (Autor et al., 2003), the response of automation investment to labor cost pressures (Acemoglu and Restrepo, 2020). Complementary evidence from cross-country robot installation data further corroborates this interpretation: Chinese robot installations nearly doubled between 2019 and 2022 while the rest of the world stagnated, with the surge concentrated in the migrant-intensive sector groups (automotive, manufacturing, and construction), while the already-robotized high-tech sectors show no additional response (Appendix A.1).

**Placebo Test** Columns (11) and (12) use total patent counts and non-automation patent counts as placebo outcomes. Both coefficients are small in magnitude and statistically indistinguishable from zero ( $-0.004$  and  $-0.005$ , respectively), a sharp contrast with the large, precisely estimated effects for automation outcomes. This asymmetry is theoretically informative: it implies that the pandemic did not broadly stimulate inventive activity in affected counties, ruling out explanations based on general increases in R&D investment or government-directed innovation subsidies during the pandemic period. Rather, the evidence points to a reallocation of innovation effort within the firm, away from product innovation and toward process automation, consistent with the substitution mechanism in the directed technical change literature (Acemoglu, 2002) and the task-based framework of Autor et al. (2003).

**Robustness: Shift Share Framework** To address the concern that pre-pandemic migrant share may be correlated with unobserved county-level trends in automation demand, we construct a shift share variable  $Z_{pt}$ . The instrument interacts each destination county’s historical migration network composition (Figure B.1) with idiosyncratic COVID-19 outflow disruptions at each origin city, measured by the AR(1) residual of the Baidu Map Migration outflow index. The construction and identifying assumptions are detailed in Appendix 5.6. Table 5 reports the reduced-form estimates. Since  $Z_{pt}$  is constructed from the residual of a mobility index, the coefficients are not directly comparable in magnitude to the baseline estimates, but the signs and significance are informative. The coefficient on  $Z_{pt}$  is negative and statistically significant across three of the four automation outcomes, consistent with the baseline DiD results. We note that the Baidu Migration data are only



available from 2019 onwards, restricting the sample relative to the baseline. Nonetheless, the results suggest that the automation response is driven by exogenous variation in COVID-19 severity at origin regions propagated through pre-existing migration networks, rather than by the endogenous sorting of migrants to high-automation counties. Appendix A.6 reports further robustness checks that replace the post-pandemic indicator with continuous exposure measures constructed from confirmed COVID-19 case counts, with consistent results.

## 5.5 Event-study

Figure 6 presents event study estimates for robot and manipulator patent shares, estimated from the following specification:

$$\text{AutoPatent}_{pt} = \alpha + \sum_{\tau \neq 0} \beta_{\tau} \left( \text{MigrantShare}_p \times \mathbf{1}[t = \tau] \right) + \gamma_p + \delta_t + X'_{p,2015} \times \delta_t + \varepsilon_{pt}, \quad (3)$$

where  $\tau$  indexes years relative to the onset of COVID-19 in 2020, which serves as the omitted reference year. The coefficients  $\beta_{\tau}$  capture the differential change in automation patenting for counties with higher pre-pandemic migrant shares in each year relative to 2020. All specifications include county fixed effects  $\gamma_p$ , province  $\times$  year fixed effects  $\delta_t$ , and interactions between initial GDP per capita and year fixed effects  $X'_{p,2015} \times \delta_t$ . Standard errors are clustered at the county level.

As illustrated in Figure 6, pre-2020 coefficients are statistically indistinguishable from zero, providing direct support for the parallel trends assumption and confirming that high-migration counties were not already on a differential automation trajectory prior to the pandemic. This is a nontrivial finding: one might have expected that counties more reliant on migrant labor would systematically differ in their innovation trajectories due to industrial composition or access to capital. The flat pre-trends mitigate this concern.

Following the 2020 outbreak, the automation response builds gradually and persistently. The effect is modest in 2021, as firms were still adjusting to the labor supply disruption and resuming operations, but expands substantially in 2022 and remains large through the end of the sample period. This temporal pattern is informative in two respects. First, it is inconsistent with the interpretation that the patent surge reflects idle engineers filing patents during shutdowns: if that were the case, we would expect the effect to be concentrated in 2020 and fade as firms reopened. Instead, the response accelerates precisely as firms resumed production and confronted the structural reality of a persistently tighter labor supply. Second, the gradual buildup suggests that automation patenting reflects a

deliberate, medium-run technological response to a sustained shift in labor availability, rather than a transitory adjustment (Hershbein and Kahn, 2018; Jaimovich and Siu, 2020). The persistence of effects beyond the immediate shock period further suggests that the automation investments embodied in patents are durable and unlikely to be reversed as labor market conditions normalize, consistent with the irreversibility of automation capital documented in Autor (2015) and Acemoglu and Restrepo (2020).

## 5.6 Bartik Instrumental Variables: Migration Networks and Migrant Inflows

We construct a Bartik instrument for county-level COVID-19-induced migrant labor supply shocks by interacting each destination county’s historical migration network (the share) with COVID-19 mobility disruptions at each origin city (the shift).<sup>12</sup> A manufacturing county that historically recruited migrants from Henan, which faced severe mobility restrictions due to its proximity to Wuhan, thus receives a more negative predicted shock than an otherwise identical county recruiting from less-affected regions. This cross-regional variation in predicted exposure, driven by origin-level epidemiological conditions rather than destination characteristics, is our identifying variation (Goldsmith-Pinkham et al., 2020).

**Historical Migration Network (Share)** We construct the migration network from bilateral flows in the 2015 China 1% Population Survey, the same source used in our baseline difference-in-differences design. For each surveyed individual we observe both registered *hukou* location (origin) and current residence (destination); aggregating and scaling by sampling weights yields the bilateral flow  $f_{op}$ , the population-level count of migrants from origin city  $o$  residing in destination county  $p$ .<sup>13</sup> The share of county  $p$ ’s migrant labor force from city  $o$  is

$$s_{op} = \frac{f_{op}}{\sum_{o'} f_{o'p}}, \quad (4)$$

so that  $\sum_o s_{op} = 1$  for all  $p$ . Shares are fixed at the 2015 baseline and thus predetermined with respect to the COVID-19 shock.

**Outflow Index (Shift)** We measure origin-city mobility disruption using daily city-level outflow data from *Baidu Migration*, a smartphone-based mobility index covering over 95%

<sup>12</sup>The instrument is constructed at the county level, an administrative division below the prefecture city and above the township in China.

<sup>13</sup>We aggregate origins to the city level to avoid the sparse-matrix problem created by many county-to-county pairs with zero observed flows (Goldsmith-Pinkham et al., 2020).

of Chinese smartphone users.<sup>14</sup> We collapse the daily index to an annual average,

$$\bar{q}_{ot} = \frac{1}{|D_t|} \sum_{d \in D_t} q_{od}, \quad (5)$$

and define the shift as the year-over-year log change,

$$g_{ot} = \log(\bar{q}_{ot}) - \log(\bar{q}_{o,t-1}). \quad (6)$$

First-differencing purges any time-invariant, city-specific level of mobility (e.g., population size, geography), leaving only the change in outflow intensity as identifying variation.<sup>15</sup> We do not residualize  $g_{ot}$  against common time effects; instead, the national component of the shock is absorbed by province-by-year fixed effects and, more formally, removed by the recentering procedure below (Borusyak and Hull, 2023). This growth-rate formulation is standard in the shift-share literature (Borusyak et al., 2022).

**Instrument Construction** Combining shares and shifts, the Bartik instrument for county  $p$  in year  $t$  is

$$Z_{pt} = \sum_o s_{op} \cdot g_{ot}. \quad (7)$$

Temporal variation in  $Z_{pt}$  comes from year-over-year fluctuations in  $g_{ot}$ , concentrated in 2020; spatial variation comes from heterogeneity in counties' historical migration networks. Two otherwise identical counties receive different predicted shocks if their origin cities were differentially disrupted by COVID-19.<sup>16</sup>

**Recentered Instrument (Borusyak–Hull Correction)** Because the shares  $s_{op}$  reflect structural, historical recruitment networks rather than random assignment,  $Z_{pt}$  has a nonzero conditional expectation  $\mu_{pt} \equiv \mathbb{E}[Z_{pt}]$  that may correlate with unobserved determinants of automation, generating an omitted-variable (“own-share”) bias.<sup>17</sup> Following Borusyak and Hull (2023), we construct  $\mu_{pt}$  directly and control for it in the reduced form, using a

<sup>14</sup>The index  $q_{od}$  measures aggregate outbound mobility intensity for city  $o$  on day  $d$ ; it does not record destinations, so origin-to-destination flows during COVID-19 are not directly observed. Destination assignment instead comes from the fixed pre-pandemic shares  $s_{op}$ .

<sup>15</sup>Baidu Migration data begin in 2019, so the predicted shock is defined from 2020 onward; large negative values of  $g_{ot}$  in 2020 reflect the unexpected contraction in outmigration from COVID-19 lockdowns.

<sup>16</sup>Identification requires that origin-city outflow disruptions affect destination automation only through migrant labor supply. This is plausible since lockdown severity was driven by local epidemiological conditions (population density, proximity to Wuhan, local policy) unrelated to destination automation decisions. First-differencing removes time-invariant origin characteristics, and recentering below removes the common national shock component.

<sup>17</sup>Identification instead relies on quasi-random assignment of the shocks  $g_{ot}$  themselves (Borusyak and Hull, 2023).

counterfactual distribution of shocks given by random within-year permutations of  $\{g_{ot}\}$  across origin cities, holding shares fixed.<sup>18</sup> Intuitively,  $\mu_{pt}$  is the predicted shock county  $p$  would have received had the same set of origin-city mobility changes been randomly reassigned across cities that year. We estimate  $\mu_{pt}$  by Monte Carlo simulation and include it as a control.<sup>19</sup>

$$\hat{\mu}_{pt} = \frac{1}{S} \sum_{s=1}^S \sum_o s_{op} \cdot g_{\pi^{(s)}(o),t} \quad (8)$$

Table 5 reports the Bartik IV estimates. Because  $Z_{pt}$  is built from a log mobility change, its coefficients are not comparable in magnitude to the baseline DiD estimates. The coefficient on  $Z_{pt}$  is negative and significant for three of five automation outcomes, mirroring the baseline results; the coefficients on robot patent counts and patents per capita are also negative, though insignificant.<sup>20</sup> These results corroborate the baseline findings: the automation response appears driven by exogenous variation in COVID-19 severity at the origin, propagated through pre-existing migration networks. Placebo coefficients on total and non-robotics patents are near zero and insignificant, further ruling out a general innovation shock. Appendix A.2 extends this design to ask whether the unpredictability of migrant labor supply, captured by a parallel Bartik IV built from lockdown frequency in the origins, independently drove the automation response.

## 6 Mechanisms and Additional Analyses

The baseline results establish that COVID-19-induced labor supply disruptions accelerated automation patenting in migrant-dependent counties. We now test the mechanism underlying this response, rule out alternative explanations, and present additional evidence on firm-level investment and worker outcomes.

### 6.1 Mechanism: Labor Market Tightness

The central premise of our identification strategy is that the COVID-19 pandemic acted as a negative labor supply shock that rendered quantity constraints binding. Economic theory predicts that firms should only substitute capital for labor when local labor reserves are

<sup>18</sup>Permuting within year preserves any aggregate national component of  $g_{ot}$  and isolates only the cross-sectional allocation of shocks as identifying variation.

<sup>19</sup>We draw  $S = 1,000$  within-year permutations  $\pi^{(s)}$  and average the resulting placebo instruments:  $\hat{\mu}_{pt} = \frac{1}{S} \sum_{s=1}^S \sum_o s_{op} \cdot g_{\pi^{(s)}(o),t}$ . This is asymptotically equivalent to the analytic leave-one-out correction in Borusyak and Hull (2023), but accommodates our unbalanced panel without closed-form influence weights;  $S = 1,000$  suffices for convergence, and larger  $S$  does not change results.

<sup>20</sup>Baidu Migration data start in 2019, and first-differencing further restricts the predicted shock to 2020 onward, tightening the estimation sample relative to baseline.

insufficient to absorb the migrant shortfall (Acemoglu, 2010; Acemoglu and Restrepo, 2018). We test this prediction through two complementary proxies for labor market tightness: the employment-to-population ratio and the pre-pandemic unemployment rate. If the automation response is driven by binding labor scarcity rather than a general pandemic shock, it should be concentrated precisely in counties where firms could not replace missing migrants with local workers.

We first define a tight labor market as a county with an employment-to-population ratio above the national median in 2015. Figure 7 presents dynamic treatment effects from an event study specification, comparing counties with tight versus slack labor markets. Prior to 2020, both groups exhibit similar trends, with coefficients fluctuating around zero, validating the parallel trends assumption. A sharp divergence emerges after the onset of COVID-19: in tight labor market counties, the treatment effect rises substantially and persists through 2023, reaching a 0.20 percentage point increase in robot-related patent share, while slack labor market counties show no discernible response throughout the post-pandemic period. The absence of any automation response in slack markets is as informative as the positive result in tight markets: it rules out the interpretation that the surge reflects a general pandemic shock rather than a binding labor supply constraint. Panel A of Table 6 confirms this in a regression framework, showing that the automation patent response is concentrated in tight labor markets while the effect in slack markets and for non-automation patents is negligible. We also examine the tight market heterogeneity using the Bartik IV and find consistent results, as illustrated in Figure 8.

## 6.2 Mechanism: County Unemployment Rate

Panel B of Table 6 splits the sample by pre-pandemic unemployment level, defining low- and high-unemployment counties by the national median. Figure 7 presents the corresponding event study estimates. As with the labor market tightness results, pre-2020 coefficients are indistinguishable from zero for both groups, supporting parallel trends. A sharp divergence emerges after 2020: low-unemployment counties exhibit a sustained increase in automation patenting, while high-unemployment counties show no discernible response. The regression results confirm this pattern: a one standard deviation increase in the migrant share (1.229 percentage points) is associated with a 0.123 percentage point increase in the robot patent share in low-unemployment counties, while the effect in high-unemployment counties is small and statistically insignificant. Non-automation patents show no significant response in either subsample.

Taken together, these two sets of results provide direct and mutually reinforcing evidence that the COVID-19 shock operated through a labor scarcity channel. Where local

labor markets were already tight and unemployment low, firms faced a binding quantity constraint when migrants became unavailable, compelling them to substitute capital for labor. Where local labor reserves were ample, the automation incentive was dampened. This pattern is precisely what directed technical change theory predicts (Acemoglu, 2010; Acemoglu and Restrepo, 2018), and it rules out broader pandemic-driven explanations for the observed automation surge.

## 6.3 Alternative Explanations

We address four alternative explanations for the observed automation response: social distancing imperatives, export demand surges, local industrial policy, and demand-side shocks. Each predicts a distinct pattern of results that is inconsistent with the data.

### 6.3.1 Alternative Explanation: Health Risk and Social Distancing

A potential concern is that firms automated not because migrant labor became unavailable, but to minimize physical contact and viral transmission risk, in which case our estimates would reflect social distancing imperatives rather than labor supply disruptions.

We introduce two proxies for pandemic-related health concerns: an industry in-person intensity index, following Dingel and Neiman (2020), that captures the employment-weighted share of jobs requiring physical presence; and 2015 population density, a proxy for transmission risk and lockdown severity.<sup>21</sup> If health concerns drove automation, these variables should absorb our main treatment effect; if labor scarcity is the operative mechanism, the migrant share interaction should remain significant regardless. We estimate the horse-race specification

$$\begin{aligned} \text{AutoPatent}_{pt} = & \alpha + \beta_1(\text{MigrantShare}_p \times \text{PostCOVID-19}_t) \\ & + \beta_2(\text{InPerson}_p \times \text{PostCOVID-19}_t) \\ & + \beta_3(\text{PopDensity}_p \times \text{PostCOVID-19}_t) + \gamma_p + \delta_t + X'_{p,2015} \times \delta_t + \varepsilon_{pt}. \end{aligned} \quad (9)$$

Columns (1)–(3) of Table 7 report the results. The migrant share interaction remains positive, significant, and stable (0.062–0.067) across all specifications, while both health-risk proxies are small and insignificant; Column (6) confirms this holds when both proxies are included alongside the export controls. These findings rule out the social distancing hypothesis: automation shows no relationship to population density or in-person work intensity, pointing instead to labor scarcity as the operative mechanism.

<sup>21</sup>Population density is total population divided by county land area in square kilometers.



### 6.3.2 Alternative Explanation: Export-oriented Automation

A second concern is that COVID-19 reshaped global demand, generating export surges that independently induced automation in export-oriented counties. China's export performance during the pandemic was indeed striking: total exports rebounded from \$2.73 trillion in 2020 to \$3.55 trillion in 2021, a roughly 30% increase, before climbing further to \$3.72 trillion in 2022 (Figure B.8).<sup>22</sup> If migrant-dependent counties are also more integrated into global value chains, failing to account for export exposure could confound our estimates.

We construct two pre-pandemic export exposure measures from 2014 customs data: a binary indicator for positive export activity (*Exporting County*) and export value as a share of county GDP (*Export Share of GDP*). Columns (4)–(5) of Table 7 introduce each measure alongside the baseline interaction; both export coefficients are small and insignificant (0.094 and  $-0.686$ , respectively), while the migrant share interaction is essentially unchanged (0.066, 0.068). Columns (6)–(7) add the social distancing proxies as well, with the same pattern.<sup>23</sup> Despite China's export boom, the automation response is concentrated in counties with the greatest migrant labor disruption rather than the greatest trade exposure, consistent with a supply-side rather than demand-side mechanism.

### 6.3.3 Alternative Explanation: Local Industrial Policy

A further concern is that local governments responded to labor shortages with automation-oriented industrial policy, so our estimates could capture top-down government-directed investment rather than firm-level responses. This concern is compounded by the fact that the central government had already been promoting automation nationwide through initiatives such as "Made in China 2025" prior to the pandemic, raising the possibility that our estimates simply reflect the continuation of pre-existing national industrial policy rather than a firm-level response to the COVID-19 labor shortage. To address this, we text-mine county government work reports for keyword frequencies across six automation-policy dimensions (robot substitution, industrial robots, automation equipment, smart manufacturing, industry upgrading, unmanned factories). As Table C.7 shows, the migrant share interaction is small and insignificant across all six dimensions and their composite, indicating no systematic policy response. The automation effect thus reflects firm-level adaptation, not top-down policy.

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<sup>22</sup>This boom was concentrated in medical supplies, electronics, and work-from-home goods, lifting China's global export market share from 12% at end-2019 to a record 15% by early 2021; net exports contributed roughly 25% of real GDP growth in 2020–2021, the largest such contribution since the late 1990s.

<sup>23</sup>This proxy captures pre-existing export exposure and cannot fully account for dynamic reallocation of export demand during the pandemic; whether trade-induced demand surges independently accelerated automation remains open for future research as richer time-varying trade data become available.

### 6.3.4 Alternative Explanation: Demand Shock

A final concern is that lockdowns reduced both labor supply and product demand simultaneously, confounding the labor scarcity interpretation. Four pieces of evidence address this. First, firm survey data show firms in high-migrant counties were less likely to have resumed operations, reported more absent workers, and were more likely to be unable to fulfill existing orders, while supply chain and logistics disruptions are uncorrelated with migrant share (Table 3)—unfilled orders are inconsistent with a demand collapse, since they indicate firms wanted to produce but could not (Lazear and Spletzer, 2012). Second, the automation response is uncorrelated with pre-pandemic export exposure (Table C.20). Third, the response is concentrated exclusively in tight, low-unemployment labor markets, where firms faced binding quantity constraints (Table 6, Figure 7)—a demand collapse would instead predict uniform effects across labor market conditions (Acemoglu, 2010; Acemoglu and Restrepo, 2018). Fourth, China’s automation surge diverged sharply from identical sectors abroad over the same period (Appendix A.1), inconsistent with a global demand shock and instead pointing to a labor supply shock specific to China’s institutional context.

Finally, a related concern is that pandemic-era R&D expenditure and firm innovation spending, rather than migrant labor scarcity, drove the automation response. We formally assess this and other omitted-variable explanations in Appendix A.7 using the sensitivity framework of Diegert et al. (2022); a confounder substantially stronger than the documented R&D expansion would be needed to overturn our central finding.

## 6.4 Demand Side: Upstream Regional Firm Investment in Automation

To shed light on the investment inputs underlying the automation patenting response, we draw on listed firm data from the China Stock Market and Accounting Research (CSMAR) database, collapsing firm-year balance sheet and R&D records to the county level.<sup>24</sup>

Table C.6 presents the results. Counties with higher pre-pandemic migrant dependence experienced significant post-COVID increases across all investment margins. A one standard deviation increase in the migrant share (1.23 percentage points) is associated with a 15.8% rise in R&D spending, an 8.9% rise in R&D personnel, a 19.5% increase in the R&D personnel ratio, and an approximate 11,730 yuan increase in fixed assets per employee. These patterns are consistent with the literature demonstrating that labor scarcity induces firms to substitute capital for labor and redirect innovation toward labor-

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<sup>24</sup>We construct annual county-level measures of R&D expenditure, R&D personnel, R&D intensity, and fixed assets per employee. The estimating equation mirrors our main specification, using the 2015 migrant population share as the treatment measure.

saving technologies (Acemoglu and Restrepo, 2018; Danzer et al., 2024; Hémous et al., 2025).

Complementary evidence from our baseline patenting results sharpens this interpretation. While automation patents respond strongly to migrant-dependent labor shortages, non-automation patenting shows no such response (Table 4). This asymmetry suggests that the post-COVID innovation response was not a broad increase in inventive activity, but was specifically directed toward labor-saving technologies—consistent with rising labor costs shifting the direction rather than the overall level of innovation (Acemoglu and Restrepo, 2018; Hémous et al., 2025). Although the CSMAR data cannot separate automation from non-automation R&D expenditures, this joint pattern suggests that labor-saving technologies were the primary target of marginal R&D spending under a tightening labor supply.<sup>25</sup>

Beyond the firm-side response, the shock carried substantial costs for the workers themselves. In Appendix A.4 we show that pre-pandemic migrant workers experienced a 4.1–4.3 percentage point increase in informal employment and income declines of 8–9 percent relative to non-migrants, with effects that persist and strengthen into 2022, well after mobility restrictions were lifted. A back-of-the-envelope aggregation of these losses, on the order of 0.8 percent of China’s 2020 GDP, is reported in Appendix A.5.

## 7 Conclusion

This paper provides causal evidence that labor scarcity induces directed technical change. Exploiting the unprecedented disruption to internal migration in China during the COVID-19 pandemic, we document a clear chain linking labor supply shocks to automation innovation.

Our analysis yields three main findings. First, the pandemic disproportionately displaced low-skilled, non-agricultural workers, reducing their cross-regional mobility relative to agricultural workers, with the disruption concentrated among those least able to work remotely and most dependent on physical mobility. Second, counties with greater pre-pandemic dependence on migrant labor experienced significant post-COVID increases in automation-related patents, with null effects on total patenting, confirming directed technical change toward labor-saving technologies rather than a generalized innovation boom (Acemoglu, 2002). The automation response is concentrated in counties with tight labor markets, where firms could not substitute missing migrants with local workers,

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<sup>25</sup>CSMAR covers only publicly listed firms, which skew toward larger establishments in major urban counties. The investment response among smaller unlisted firms—which employ the bulk of migrants—remains unobserved, so these results should be interpreted as suggestive given potential selection concerns in the listed-firm sample.

providing direct evidence that binding and recurrent quantity constraints rather than price changes are the operative mechanism (Acemoglu, 2010). Third, we trace the distributional consequences of this technological transition: displaced migrant workers experienced significant increases in informal employment and persistent income losses, bearing the bulk of the adjustment costs, comparable in incidence to the distributional costs of robot adoption in the United States (Acemoglu and Restrepo, 2020) and China’s trade liberalization (Autor et al., 2013).

Several alternative explanations are ruled out. The automation response is not driven by social distancing imperatives, export demand surges, or local government industrial policy, but reflects firm-level adaptation to binding labor supply constraints.

Our findings suggest that transient mobility frictions can trigger permanent industrial upgrading. More broadly, we demonstrate that sudden, high-frequency shocks can redirect the direction of innovation when labor constraints become acute, with lasting consequences for both technological trajectories and the distribution of gains and losses across workers.

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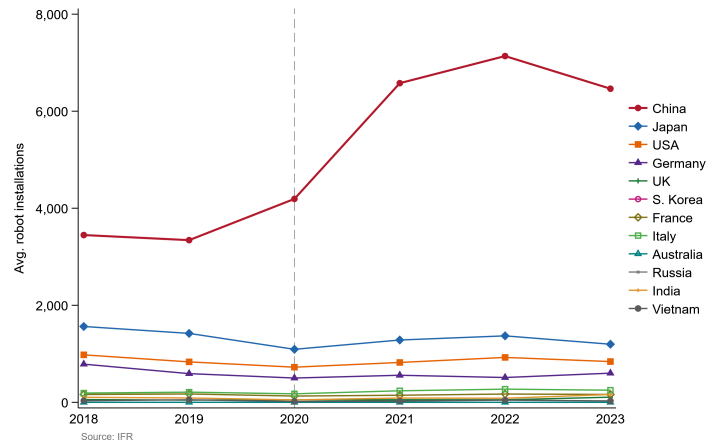
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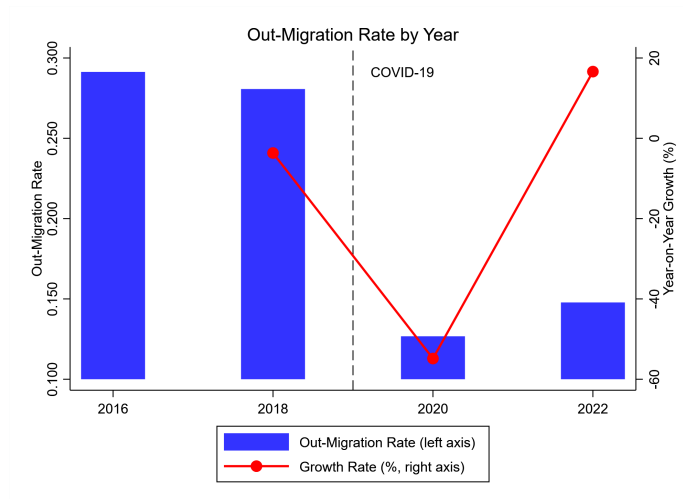
## 8 Figures

Figure 1: Robotics Installation by Country Before and After COVID-19



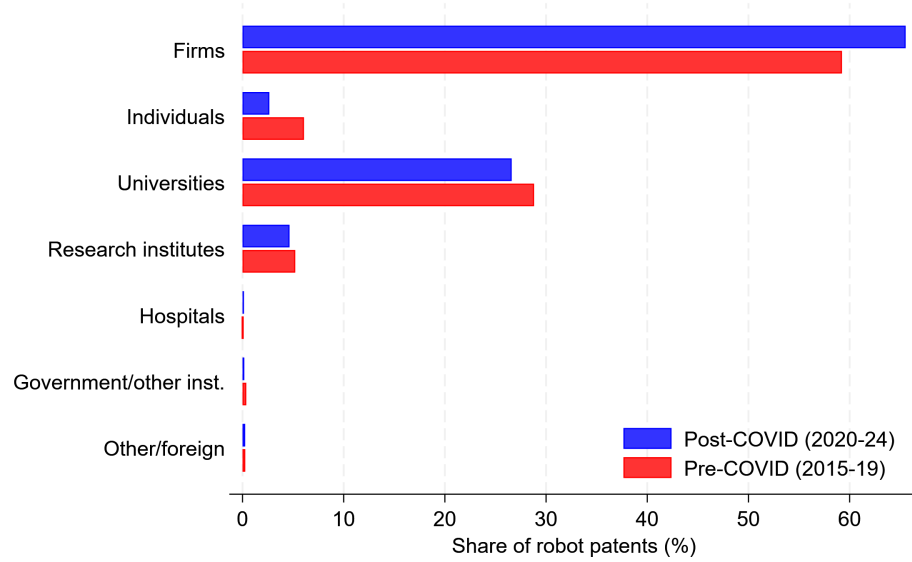
Notes: Sourced from International Federation of Robotics Reports <https://ifr.org/worldrobotics>

Figure 2: Out-migration Rate by Year: Before and After COVID-19



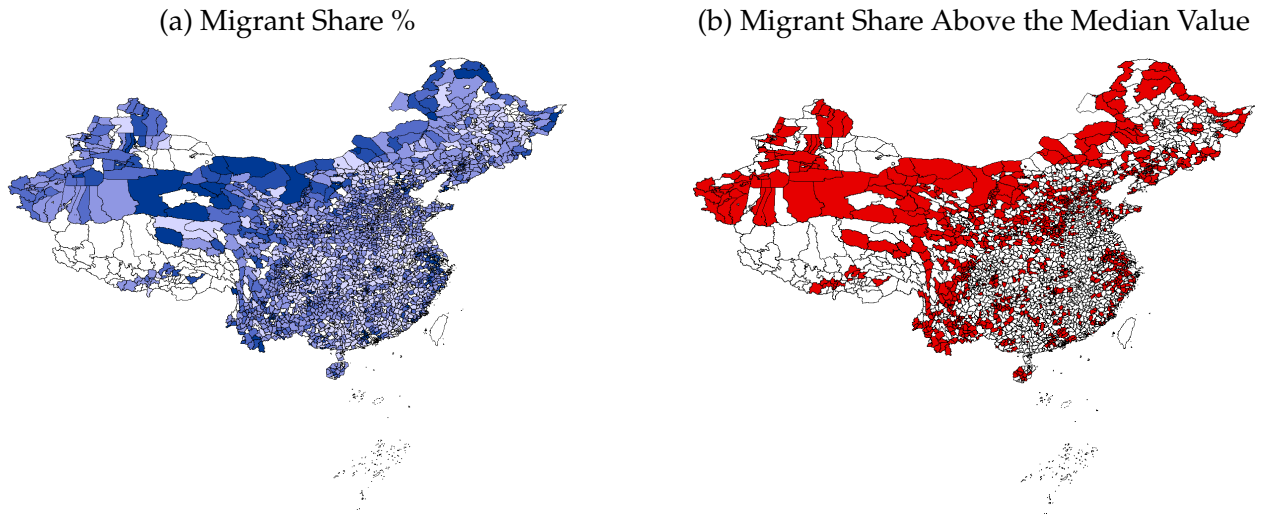
Notes: This figure displays the average out-migration/migrant worker rate by year using individual survey data (CFPS).

Figure 3: Firms File Patents after COVID-19



Notes: This figure plots the share of firms filing robotics/automation patents before and after COVID-19.

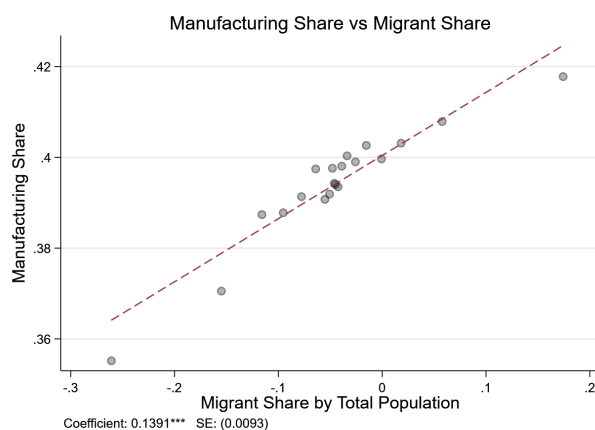
Figure 4: Regional Treatment Intensity and Treatment Indicator in 2015



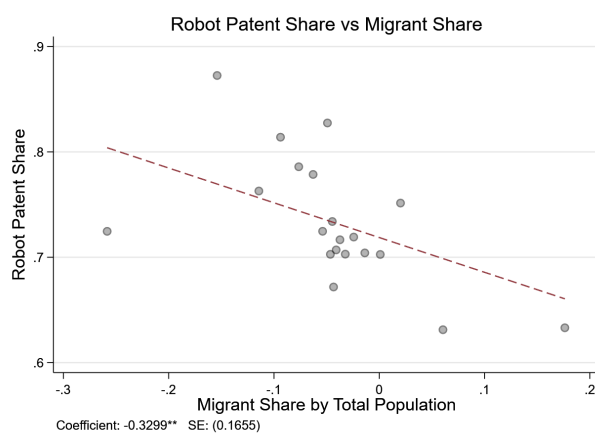
Notes: This figure displays regional treatment intensity and treatment status across Chinese counties in 2015. Panel (a) shows the continuous migrant share, defined as the number of migrants relative to the total population in each county. Panel (b) shows the binary treatment indicator, which equals one if a county's migrant share exceeds the sample median. Data are drawn from the 2015 China 1% Population Survey (mini-census).

Figure 5: Migrant Share and Manufacturing/Automation Before COVID-19

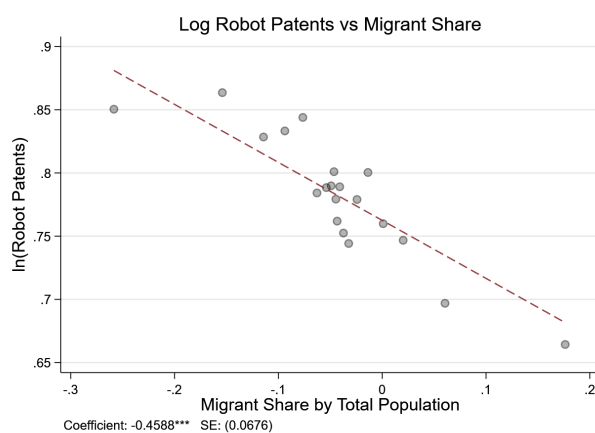
(a) Manufacturing Output Share by GDP



(b) Robot Patent Share

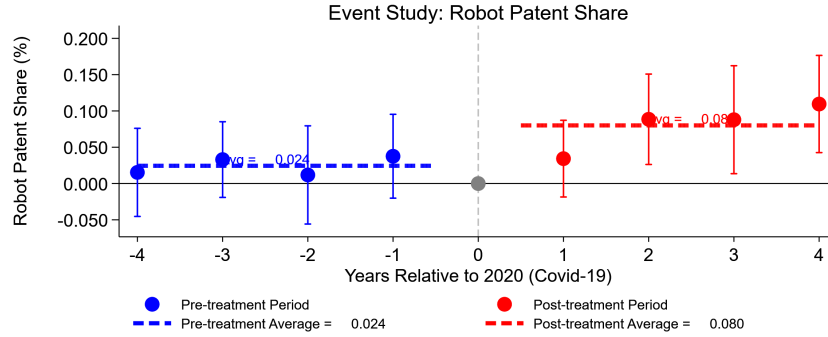


(c) Robot Patent Count



Notes: Each panel shows the binscatter plot with county fixed effects absorbed. The sample period is 2015-2019 (prior to COVID-19). Coefficients and standard errors are from reghdfe regressions with county fixed effects and clustered standard errors at the county level. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

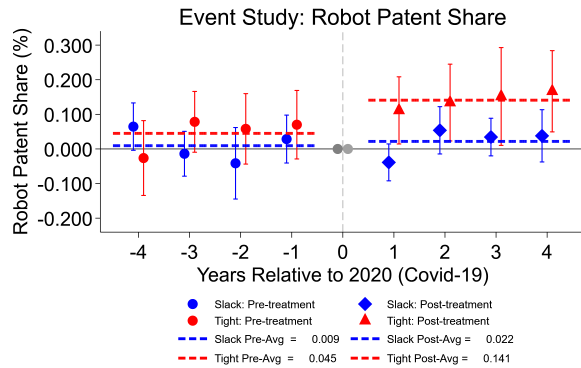
Figure 6: Event Study for County Robotics Patent %



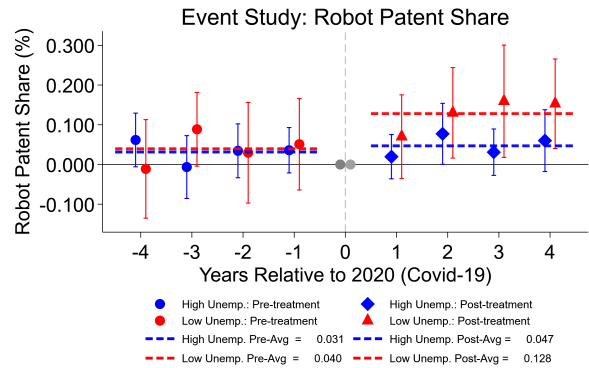
*Notes:* This figure plots dynamic treatment effects from an event study specification. The dependent variables are the share of robotics patents. The treatment variable is the pre-pandemic number of migrants relative to the total population in each county. The omitted reference year is 2020. All specifications include county fixed effects and interactions between initial conditions (GDP per capita in 2015) and province-year fixed effects. Vertical bars represent 95% confidence intervals based on standard errors clustered at the county level. The vertical dashed line indicates the onset of COVID-19 in 2020.

Figure 7: Tight vs. Slack Market Response: Event Study for County Robotics Patent %

(a) Counties with Tight Market vs. Slack Market Before COVID-19

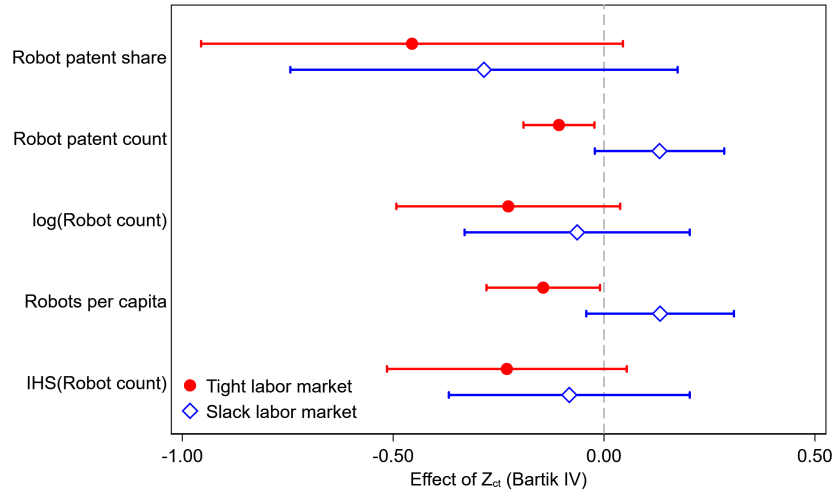


(b) Counties with Low Unemployment Rate vs. High Unemployment Rate Before COVID-19



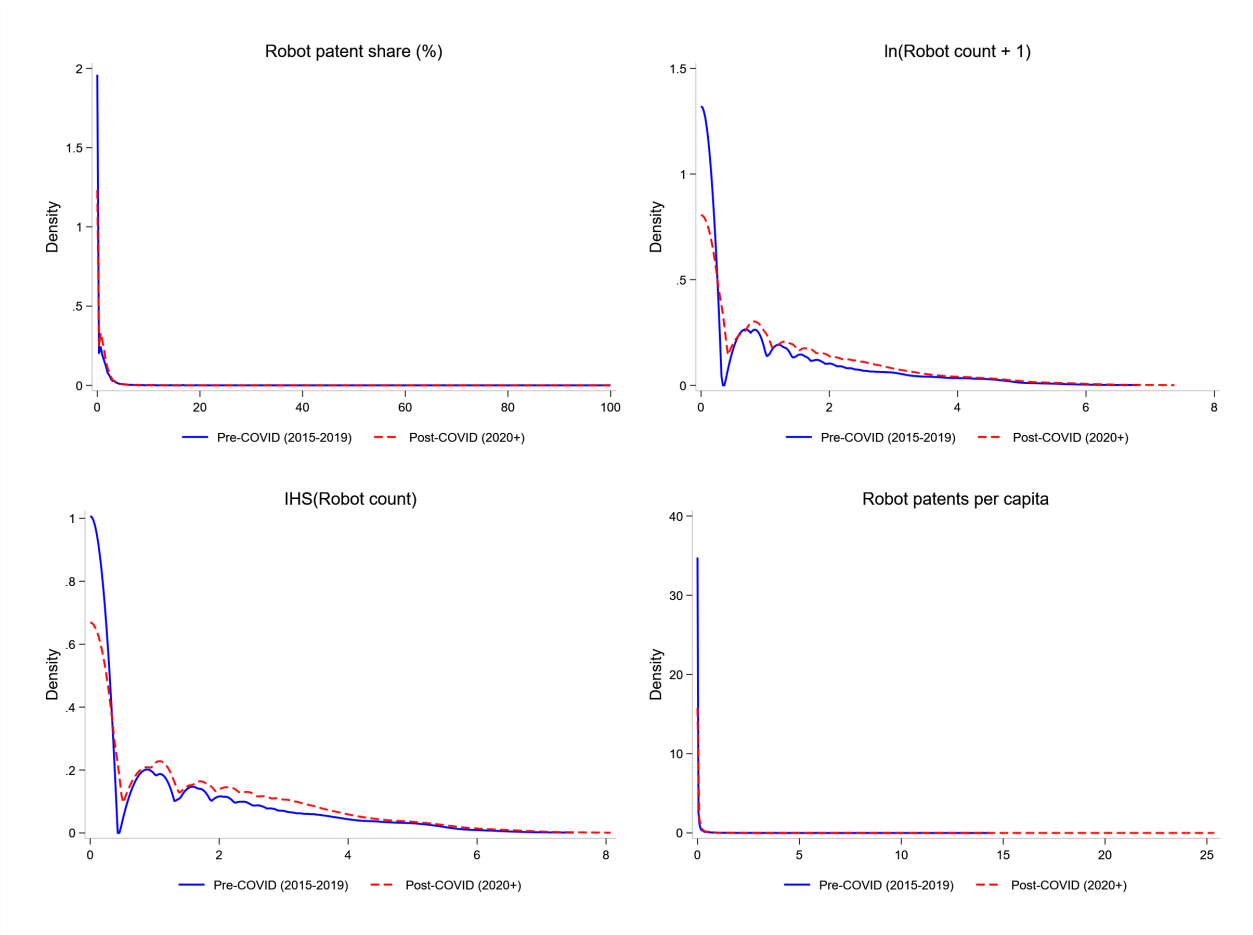
*Notes:* This figure plots dynamic treatment effects from an event study specification, examining heterogeneous responses by local labor market conditions. The dependent variable is the share of robotics patents among total patents in each county. Panel (a) compares counties with tight labor markets versus slack labor markets prior to COVID-19. Panel (b) compares counties with low unemployment rates versus high unemployment rates prior to COVID-19. The omitted reference year is 2020. All specifications include county fixed effects and interactions between initial conditions (GDP per capita in 2015) and province-year fixed effects. Vertical bars represent 95% confidence intervals based on standard errors clustered at the county level. The vertical dashed line indicates the onset of COVID-19 in 2020. Horizontal dashed lines show the average post-treatment effects for each group. Points are horizontally offset for visual clarity.

Figure 8: Tight vs. Slack Market Response to Disruptions in Origins



*Notes:* This figure plots Bartik IV coefficients on robot patent share for counties split by pre-pandemic labor market tightness. The instrument  $Z_{pt}$  is the shift-share migrant labor supply shock described in Section 5.6, controlling for the recentering term  $\mu_{pt}$ . Tight (Slack) counties are those with above- (below-) median labor market tightness prior to COVID-19. All specifications include county fixed effects, province  $\times$  year fixed effects, and initial GDP per capita  $\times$  year fixed effects. Vertical bars represent 95% confidence intervals based on standard errors clustered at the county level.

Figure 9: Distribution of Automation Patents by Periods: Changes in Extensive Margin



*Notes:* This figure plots the kernel density distributions of automation patent variables before and after COVID-19. The top panels show log counts of robot patents (left) and manipulator patents (right). The bottom panels show the share of robot patents (left) and manipulator patents (right) among total patents. Pre-COVID period: 2015–2019; Post-COVID period: 2020–2024. Data sourced from the China National Intellectual Property Administration (CNIPA).



## 9 Tables

Table 1: Summary Statistics of Patent Variables by Migration Status and Period

|                            | High Migration County |       |       |        | Low Migration County |       |       |       |
|----------------------------|-----------------------|-------|-------|--------|----------------------|-------|-------|-------|
|                            | Mean                  | SD    | Min   | Max    | Mean                 | SD    | Min   | Max   |
| <i>Panel A: Pre-COVID</i>  |                       |       |       |        |                      |       |       |       |
| Robot patent share (%)     | 0.845                 | 2.390 | 0     | 82.609 | 0.683                | 2.257 | 0     | 100   |
| ln(Robot count + 1)        | 1.105                 | 1.483 | 0     | 6.807  | 0.538                | 0.871 | 0     | 5.529 |
| IHS(Robot count)           | 1.344                 | 1.742 | 0     | 7.499  | 0.681                | 1.077 | 0     | 6.219 |
| Robot patents per capita   | 0.204                 | 0.663 | 0     | 14.333 | 0.036                | 0.151 | 0     | 4.841 |
| ln(Total patents + 1)      | 4.972                 | 2.062 | 0.693 | 10.547 | 4.441                | 1.467 | 0.693 | 9.657 |
| ln(Non-auto patents + 1)   | 4.937                 | 2.057 | 0     | 10.508 | 4.407                | 1.465 | 0     | 9.618 |
| <i>Panel B: Post-COVID</i> |                       |       |       |        |                      |       |       |       |
| Robot patent share (%)     | 0.904                 | 1.296 | 0     | 20     | 0.750                | 2.391 | 0     | 100   |
| ln(Robot count + 1)        | 1.409                 | 1.608 | 0     | 7.377  | 0.781                | 0.987 | 0     | 5.204 |
| IHS(Robot count)           | 1.706                 | 1.870 | 0     | 8.070  | 0.985                | 1.215 | 0     | 5.892 |
| Robot patents per capita   | 0.331                 | 1.057 | 0     | 25.365 | 0.054                | 0.155 | 0     | 4.583 |
| ln(Total patents + 1)      | 5.398                 | 2.024 | 0.693 | 10.838 | 4.871                | 1.487 | 0.693 | 9.433 |
| ln(Non-auto patents + 1)   | 5.367                 | 2.018 | 0     | 10.770 | 4.840                | 1.486 | 0     | 9.409 |

*Notes:* This table reports summary statistics for patent variables by migration status and time period. Panel A covers the pre-COVID period and Panel B covers the post-COVID period. High and Low migration groups are defined by the median net migrant share in 2015.

Table 2: Impact of COVID-19 on Migration by Pre-Pandemic Employment Sector

| VARIABLES                                                  | Out-of-Hometown Work |                      |                      |
|------------------------------------------------------------|----------------------|----------------------|----------------------|
|                                                            | (1)                  | (2)                  | (3)                  |
| Non-Agricultural Worker <sub>pre-COVID</sub><br>× COVID-19 | -0.095***<br>(0.007) | -0.111***<br>(0.008) | -0.110***<br>(0.008) |
| Observations                                               | 37,377               | 36,880               | 37,239               |
| R-squared                                                  | 0.635                | 0.457                | 0.640                |
| Individual FEs                                             | ✓                    |                      | ✓                    |
| Household FEs                                              |                      | ✓                    |                      |
| Year FEs                                                   | ✓                    | ✓                    |                      |
| Province × Year FEs                                        |                      |                      | ✓                    |

*Notes:* The dependent variable is an indicator for out-of-hometown migrant work. The key coefficient captures the interaction between pre-pandemic non-agricultural employment status and a post-2020 indicator (COVID-19 period). Columns differ in the fixed effects included: Column (1) includes individual and year fixed effects; Column (2) includes household and year fixed effects; Column (3) includes individual and province-by-year fixed effects. Standard errors clustered at the individual level are reported in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 3: Correlation between Migrant Labor Dependence and Firm-level Disruptions During COVID-19 (in 2020)

| VARIABLES              | 1(Resumed Operations)<br>(1) | Absent Workers Share<br>(2) | 1(No Delivered Orders)<br>(3) | 1(Wage Increase for Labor)<br>(4) | 1(Labor Cost Pressure)<br>(5) | 1(Protection Shortage)<br>(6) | 1(Supply Chain Disruption)<br>(7) | 1(Logistics Disruption)<br>(8) |
|------------------------|------------------------------|-----------------------------|-------------------------------|-----------------------------------|-------------------------------|-------------------------------|-----------------------------------|--------------------------------|
| Firm Migrant Worker %  | −0.127***<br>(0.038)         | 5.198**<br>(2.189)          | 0.057*<br>(0.032)             | 0.057<br>(0.043)                  | −0.049<br>(0.048)             | −0.009<br>(0.047)             | −0.005<br>(0.029)                 | −0.034<br>(0.050)              |
| Observations (# Firms) | 942                          | 798                         | 942                           | 942                               | 942                           | 942                           | 942                               | 942                            |
| R-squared              | 0.219                        | 0.192                       | 0.115                         | 0.148                             | 0.243                         | 0.131                         | 0.133                             | 0.202                          |
| Firm Characteristics   | ✓                            | ✓                           | ✓                             | ✓                                 | ✓                             | ✓                             | ✓                                 | ✓                              |
| City FEs               | ✓                            | ✓                           | ✓                             | ✓                                 | ✓                             | ✓                             | ✓                                 | ✓                              |
| Industry FEs           | ✓                            | ✓                           | ✓                             | ✓                                 | ✓                             | ✓                             | ✓                                 | ✓                              |

*Notes:* This table examines the relationship between migrant labor dependence and firm-level operational disruptions during the COVID-19 pandemic, using firm survey data collected in early 2020 (ESIEC). Each column reports an OLS regression where the dependent variable captures a distinct dimension of firm disruption: whether the firm had resumed operations (Column 1), the share of workers absent due to mobility restrictions (Column 2), inability to fulfill existing orders (Column 3), whether the firm reported wage increases (Column 4), labor cost pressures (Column 5), shortages of personal protective equipment (Column 6), supply chain disruptions (Column 7), and logistics disruptions (Column 8). Migrant Share is the firm migrant worker share in 2017. All specifications include city fixed effects, industry fixed effects, and firm characteristics. Standard errors clustered at the city level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 4: Baseline: Effect of Migration Restrictions on Automation Adoption

| VARIABLES                                                                                          | Automation-Related Patent |          |                    |          |                   |          |                  |          |                  |          | Placebo            |                       |
|----------------------------------------------------------------------------------------------------|---------------------------|----------|--------------------|----------|-------------------|----------|------------------|----------|------------------|----------|--------------------|-----------------------|
|                                                                                                    | Robot Patent Share %      |          | Robot Patent Count |          | ln(Robot Count+1) |          | Robot per Capita |          | IHS(Robot Count) |          | ln(Total Patent+1) | ln(Non-Auto Patent+1) |
|                                                                                                    | (1)                       | (2)      | (3)                | (4)      | (5)               | (6)      | (7)              | (8)      | (9)              | (10)     | (11)               | (12)                  |
| <i>Panel A: County FEs + Year FEs</i>                                                              |                           |          |                    |          |                   |          |                  |          |                  |          |                    |                       |
| Migrant Share % $\times$ COVID-19                                                                  | 0.049***                  |          | 6.814***           |          | 0.051***          |          | 0.093***         |          | 0.051***         |          |                    |                       |
|                                                                                                    | (0.015)                   |          | (1.395)            |          | (0.010)           |          | (0.016)          |          | (0.011)          |          |                    |                       |
| <i>Panel B: + Province <math>\times</math> Year FEs + Initial GDP <math>\times</math> Year FEs</i> |                           |          |                    |          |                   |          |                  |          |                  |          |                    |                       |
| Migrant Share % $\times$ COVID-19                                                                  |                           | 0.063*** |                    | 5.457*** |                   | 0.035*** |                  | 0.081*** |                  | 0.034*** | -0.004             | -0.005                |
|                                                                                                    |                           | (0.021)  |                    | (1.231)  |                   | (0.010)  |                  | (0.015)  |                  | (0.012)  | (0.010)            | (0.010)               |
| Dep. Var. Mean                                                                                     | 0.792                     | 0.792    | 10.211             | 10.211   | 0.944             | 0.944    | 0.151            | 0.151    | 1.163            | 1.163    | 4.910              | 4.877                 |
| No. of Counties                                                                                    | 2,165                     | 2,165    | 2,165              | 2,165    | 2,165             | 2,165    | 2,165            | 2,165    | 2,165            | 2,165    | 2,165              | 2,165                 |
| Observations                                                                                       | 21,568                    | 21,568   | 21,568             | 21,568   | 21,568            | 21,568   | 21,568           | 21,568   | 21,568           | 21,568   | 21,568             | 21,568                |
| R-squared                                                                                          | 0.194                     | 0.219    | 0.789              | 0.826    | 0.841             | 0.863    | 0.791            | 0.822    | 0.829            | 0.850    | 0.943              | 0.942                 |
| County FEs                                                                                         | ✓                         | ✓        | ✓                  | ✓        | ✓                 | ✓        | ✓                | ✓        | ✓                | ✓        | ✓                  | ✓                     |
| Year FEs                                                                                           | ✓                         |          | ✓                  |          | ✓                 |          | ✓                |          | ✓                |          |                    |                       |
| Province $\times$ Year FEs                                                                         |                           | ✓        |                    | ✓        |                   | ✓        |                  | ✓        |                  | ✓        | ✓                  | ✓                     |
| Initial GDP $\times$ Year FEs                                                                      |                           | ✓        |                    | ✓        |                   | ✓        |                  | ✓        |                  | ✓        | ✓                  | ✓                     |

*Notes:* This table reports the effect of labor mobility restrictions on automation adoption across alternative fixed effect specifications. The key independent variable is the interaction between county-level migrant share and COVID-19 onset. Columns (1)–(2), (3)–(4), (5)–(6), (7)–(8), and (9)–(10) report results for robot patent share, robot patent count, log robot count, robots per capita, and IHS-transformed robot count, respectively. Columns (11) and (12) use total patents and non-automation patents as placebo dependent variables. Panel A includes county and year fixed effects only; Panel B adds Province  $\times$  Year and Initial GDP  $\times$  Year fixed effects. Standard errors clustered at the county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 5: Bartik Instrumental Variable: Origin Disruptions and Destination Robot Adoption

| VARIABLES                                            | Automation-Related Patent |                   |                    |                   |                    | Placebo            |                       |
|------------------------------------------------------|---------------------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------------|
|                                                      | Robot Patent Share %      | Robot Count       | ln(Robot Count+1)  | Robot per Capita  | IHS(Robot Count)   | ln(Total Patent+1) | ln(Non-Auto Patent+1) |
|                                                      | (1)                       | (2)               | (3)                | (4)               | (5)                | (6)                | (7)                   |
| $Z_{p,t}$ (Shift Share Variable)                     | -0.679**<br>(0.325)       | -3.237<br>(2.293) | -0.218*<br>(0.130) | -0.057<br>(0.041) | -0.272*<br>(0.164) | 0.024<br>(0.099)   | 0.024<br>(0.100)      |
| $\mu_t$ (Recentering Term, $E[Z]$ )                  | 1.935<br>(1.634)          | 5.415<br>(3.284)  | 0.345<br>(0.278)   | 0.100<br>(0.068)  | 0.430<br>(0.352)   | -0.139<br>(0.255)  | -0.157<br>(0.256)     |
| Observations                                         | 13,124                    | 13,124            | 13,124             | 13,124            | 13,124             | 13,124             | 13,124                |
| R-squared                                            | 0.277                     | 0.892             | 0.896              | 0.883             | 0.884              | 0.962              | 0.961                 |
| County FEs                                           | ✓                         | ✓                 | ✓                  | ✓                 | ✓                  | ✓                  | ✓                     |
| Province $\times$ Year FEs                           | ✓                         | ✓                 | ✓                  | ✓                 | ✓                  | ✓                  | ✓                     |
| Initial GDP $\times$ Year FEs                        | ✓                         | ✓                 | ✓                  | ✓                 | ✓                  | ✓                  | ✓                     |
| <b>Borusyak and Hull (2023)</b> Recentering Controls | ✓                         | ✓                 | ✓                  | ✓                 | ✓                  | ✓                  | ✓                     |

*Notes:* This table reports Bartik IV estimates of the effect of COVID-19-induced migrant labor supply shocks on automation adoption. The sample spans 2019–2024, reflecting the availability of Baidu Migration outflow data from 2019 onward.  $Z_{p,t}$  is the shift-share instrument, constructed as the share-weighted sum of origin-city outflow shocks: the share is each destination county's pre-pandemic migration network composition from the 2015 China Population Census, and the shift is the year-over-year log change in each origin city's Baidu Migration outflow index.  $\mu_t$  is the recentering term of **Borusyak and Hull (2023)**, estimated via Monte Carlo simulation as the expected instrument value under random within-year reassignment of shifts across origin cities, and included in every specification to correct for non-random exposure shares. Columns (6) and (7) report placebo outcomes, total patents and non-automation patents, which should be unaffected if the exclusion restriction holds. All columns include county fixed effects, province  $\times$  year fixed effects, and initial GDP per capita  $\times$  year fixed effects. Standard errors clustered at the county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 6: Heterogeneity Analyses: Labor Market Tightness and Unemployment

| VARIABLES                         | Robot Patent Share  |                  | ln(Non-Auto Patent+1) |                   | Robot Patent Share  |                    | ln(Non-Auto Patent+1) |                    |
|-----------------------------------|---------------------|------------------|-----------------------|-------------------|---------------------|--------------------|-----------------------|--------------------|
|                                   | Tight<br>(1)        | Slack<br>(2)     | Tight<br>(3)          | Slack<br>(4)      | Low Unemp.<br>(5)   | High Unemp.<br>(6) | Low Unemp.<br>(7)     | High Unemp.<br>(8) |
| Migrant Share % $\times$ COVID-19 | 0.103***<br>(0.036) | 0.015<br>(0.023) | -0.010<br>(0.015)     | -0.004<br>(0.014) | 0.100***<br>(0.034) | 0.025<br>(0.023)   | -0.017<br>(0.015)     | 0.005<br>(0.012)   |
| Observations                      | 11,266              | 10,265           | 11,266                | 10,265            | 10,445              | 10,427             | 10,445                | 10,427             |
| R-squared                         | 0.228               | 0.216            | 0.945                 | 0.940             | 0.214               | 0.296              | 0.920                 | 0.954              |
| County FEs                        | ✓                   | ✓                | ✓                     | ✓                 | ✓                   | ✓                  | ✓                     | ✓                  |
| Province $\times$ Year FEs        | ✓                   | ✓                | ✓                     | ✓                 | ✓                   | ✓                  | ✓                     | ✓                  |
| Initial GDP $\times$ Year FEs     | ✓                   | ✓                | ✓                     | ✓                 | ✓                   | ✓                  | ✓                     | ✓                  |

*Notes:* This table reports split-sample heterogeneity analyses for the effect of COVID-19 on automation patents. Columns 1–4 split the sample by labor market tightness (tight vs. slack markets); Columns 5–8 split by unemployment level (low vs. high unemployment). The dependent variables are the robot patent share (Columns 1–2 and 5–6) and log non-automation patents (Columns 3–4 and 7–8). All specifications include county fixed effects, province  $\times$  year fixed effects, and initial GDP  $\times$  year fixed effects. Standard errors clustered at the county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 7: Alternative Explanations: Industry In-Person Intensity, Population Density, and Export Exposure

| VARIABLES                                      | Robot Patent Share  |                    |                     |                     |                     |                    |                    |
|------------------------------------------------|---------------------|--------------------|---------------------|---------------------|---------------------|--------------------|--------------------|
|                                                | (1)                 | (2)                | (3)                 | (4)                 | (5)                 | (6)                | (7)                |
| Migrant Share % $\times$ COVID-19              | 0.063***<br>(0.021) | 0.062**<br>(0.024) | 0.067***<br>(0.024) | 0.066***<br>(0.024) | 0.068***<br>(0.024) | 0.061**<br>(0.024) | 0.062**<br>(0.025) |
| Industry In-Person Intensity $\times$ COVID-19 |                     | 0.003<br>(0.004)   |                     |                     |                     | 0.002<br>(0.004)   | 0.002<br>(0.004)   |
| Population Density $\times$ COVID-19           |                     |                    | 0.252<br>(0.249)    |                     |                     | 0.214<br>(0.262)   | 0.243<br>(0.253)   |
| Exporting County $\times$ COVID-19             |                     |                    |                     | 0.094<br>(0.089)    |                     | 0.092<br>(0.088)   |                    |
| Export Share of GDP $\times$ COVID-19          |                     |                    |                     |                     | -0.686<br>(1.167)   |                    | -0.700<br>(1.169)  |
| Observations                                   | 21,568              | 18,723             | 18,723              | 18,723              | 18,723              | 18,723             | 18,723             |
| R-squared                                      | 0.219               | 0.234              | 0.234               | 0.234               | 0.233               | 0.234              | 0.233              |
| County FEs                                     | ✓                   | ✓                  | ✓                   | ✓                   | ✓                   | ✓                  | ✓                  |
| Province $\times$ Year FEs                     | ✓                   | ✓                  | ✓                   | ✓                   | ✓                   | ✓                  | ✓                  |
| Initial GDP $\times$ Year FEs                  | ✓                   | ✓                  | ✓                   | ✓                   |                     | ✓                  |                    |

*Notes:* This table tests whether the COVID-19-induced automation response was driven by industry in-person intensity, population density, or export exposure rather than labor scarcity. The dependent variable is robot patent share. Migrant Share is net migration as a share of total population in 2015. Industry In-Person Intensity is the employment-weighted average of industry-level inability-to-work-from-home shares from Dingel and Neiman (2020), computed using each county's pre-period industry employment distribution; higher values indicate a county's workforce is more concentrated in industries requiring on-site presence. Population Density is population per square kilometer in 2015. Exporting County is a binary indicator for counties with positive export activity in the pre-period. Export Share of GDP is the value of exports as a share of county GDP in the pre-period. COVID-19 indicates years 2020–2023. Column (1) uses the full baseline sample; Columns (2)–(7) are restricted to the common sample across all alternative control specifications. All specifications include county fixed effects, province  $\times$  year fixed effects, and interactions of initial GDP per capita with year fixed effects. Standard errors clustered at county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .



# Online Appendix

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## A Additional Results

### A.1 Robotics Installation: China vs. Rest of the World

To distinguish domestic labor supply shocks from global industry trends, we complement our analysis of indigenous innovation with data on physical capital adoption. We obtain industrial robot installation statistics from the International Federation of Robotics (IFR), covering the period from 2014 to 2023. While our patent data captures the development of technology within China, the global scope of the IFR database allows us to construct a necessary counterfactual: tracking how automation evolved in the rest of the world during the same period. This source is widely recognized as the authoritative standard for cross-country comparisons and has been extensively employed in the literature (Graetz and Michaels, 2018; Acemoglu and Restrepo, 2020).

The IFR data report annual robot installations (newly deployed industrial robots) disaggregated by country and industry sector. Industrial robots are defined as automatically controlled, reprogrammable, multipurpose manipulators programmable in three or

more axes, which can be either fixed in place or mobile for use in industrial automation applications. The data exclude service robots and focus exclusively on robots used in manufacturing and related industrial activities. Each installation represents a unit deployment, regardless of the robot’s capabilities or price.

For our analysis, we focus on several key dimensions of the robot installation data. First, we construct country-level measures of robot adoption, with particular attention to China versus the rest of the world. China has emerged as the world’s largest market for industrial robots, accounting for over 50% of global installations since 2016. Second, we aggregate the detailed IFR industries into five sector groups that differ in their reliance on low-skilled internal migrant labor in China: manufacturing and construction, automotive, labor-intensive manufacturing, high-tech, and electronics and electrical.<sup>26</sup> This grouping allows us to test whether COVID-19 had differential effects on automation across industries with different exposure to the migrant labor supply shock.

We construct our primary dependent variable as  $\ln(\text{Robot Installations} + 1)$  to account for zero installations in some country-sector-year cells and to facilitate interpretation of coefficients as semi-elasticities. We also use the current stocks of robotics as the secondary outcome variable. Our final dataset includes robot installation data for over 100 countries and 30 detailed sectors.<sup>27</sup>

We employ a difference-in-differences framework augmented with a triple interaction to capture sector-group heterogeneity. The estimating equation is:

$$Y_{cst} = \alpha + \beta(\text{Group}_s \times \text{China}_c \times \text{PostCOVID-19}_t) + \gamma_c + \lambda_s + \delta_t + \varepsilon_{cst}, \quad (10)$$

where  $Y_{cst}$  denotes the log of new robot installations in country  $c$ , industry sector  $s$ , and year  $t$ . The term  $\text{Group}_s$  is a binary indicator for one of the five sector groups defined above,  $\text{China}_c$  identifies the domestic economy, and  $\text{PostCOVID-19}_t$  takes the value of one for years 2020 and after.

The specification includes a set of fixed effects to absorb invariant heterogeneity:  $\gamma_c$

---

<sup>26</sup>The groups aggregate the detailed IFR industries as follows. *Manufacturing and construction* comprises all industries except agriculture, forestry and fishing, electricity, gas and water supply, and education/research/development. *Automotive* comprises motor vehicles, engines and bodies, other vehicles, and all automotive parts (metal, glass, electrical/electronic, rubber and plastic, and other). *Labor-intensive manufacturing* comprises food and beverages, textiles, wood and furniture, and paper. *High-tech* comprises semiconductors, LCD and LED, electronic components and devices, computers and peripheral equipment, information and communication equipment, medical, precision and optical instruments, and pharmaceuticals and cosmetics. *Electronics and electrical* comprises semiconductors, LCD and LED, electronic components and devices, computers and peripheral equipment, information and communication equipment, electrical machinery, and household appliances. The high-tech and electronics and electrical groups overlap in four industries and are therefore not independent tests. Groups are not mutually exclusive; each regression compares one group against all remaining industries.

<sup>27</sup>We also use robotics stock as the second outcome variable for robustness.

captures country fixed effects;  $\lambda_s$  accounts for sector fixed effects; and  $\delta_t$  denotes year fixed effects. The coefficient of interest,  $\beta$ , captures the differential acceleration of robot adoption in each of China's sector groups. We expect positive coefficients for the migrant-intensive groups: according to official data, the manufacturing and construction sectors alone absorb nearly half of China's total migrant workforce ([National Bureau of Statistics of China, 2021](#)). The high-tech and electronics groups provide a useful contrast: [Cheng et al. \(2019\)](#) document that the electronics and automotive industries have historically possessed the highest labor intensity, making them early targets for robotization as labor costs rose, so much of their automation predates the pandemic. If the post-2020 surge instead reflected a technology-push or subsidy-driven expansion, it should load on these already-advanced groups rather than on the labor-reliant ones.

Table [C.16](#) presents difference-in-differences estimates for robot installations on a global sample of countries. Panel A reports the baseline China  $\times$  COVID-19 coefficient, capturing the aggregate differential increase in robot adoption in China relative to the rest of the world. Panel B reports the triple-interaction terms (China  $\times$  COVID-19  $\times$  Group), isolating the additional surge specific to each sector group within China.

The results indicate a substantial and country-specific pivot toward automation. Column (1) shows that after 2020, robot installations in China rose by roughly 1.2 log points relative to the rest of the world, a differential of more than 200 percent, confirming that Chinese manufacturers accelerated automation far more rapidly than their global peers. Panel B reveals that this acceleration is concentrated precisely in the groups most reliant on migrant labor. The additional effect is largest for the automotive group (0.78 log points) and the broad manufacturing and construction group (0.71), with a smaller but significant differential in labor-intensive manufacturing (0.21). The negative China  $\times$  Group level terms for these groups indicate that they entered the pandemic relatively under-automated, so the surge represents rapid catch-up investment exactly where production still relied on physical labor. By contrast, the high-tech and electronics and electrical groups, in which China was already heavily robotized before 2020 (their China  $\times$  Group level terms are large and positive), show significantly negative additional effects. This asymmetry doubles as a falsification test of the innovation-push alternative: a subsidy-driven or technology-push expansion would concentrate where China's robotics capabilities and industrial-policy support were already strongest, namely high-tech and electronics, whereas the observed response loads exclusively on the sectors where migrant labor became scarce. The pandemic did not amplify an existing high-tech specialization; it redirected automation investment toward previously labor-reliant activities. Meanwhile, Figure [1](#) shows that installations in the rest of the world experienced muted or negative trends during the pandemic. The divergence confirms that the impetus for automation was not a global shift in final product

demand, but a shock to labor availability specific to the Chinese institutional context.

To ensure the main results reflect structural capital deepening rather than a temporary investment spike, we replicate the analysis using operational robot stocks as the outcome variable. These results are reported in Table C.17. The aggregate effect is confirmed: China's operational robot stock rose by 0.96 log points relative to the rest of the world after 2020, indicating that the installation surge accumulated into a lasting expansion of the capital stock rather than a one-off purchase spike. At the group level, the stock differentials are more muted than the installation differentials, as stocks embed pre-pandemic vintages and adjust slowly to new investment flows; the labor-intensive group nonetheless shows a significantly positive stock differential (0.31 log points). Because the installation flow directly measures the pandemic-era investment decision, we treat the installation estimates as the primary evidence on sectoral reallocation and the stock estimates as evidence on its persistence. Figure B.7 complements the regression evidence descriptively, displaying the composition of China's operational robot stock across the focal sectors before and after COVID-19; the post-pandemic expansion of the construction share is consistent with the catch-up pattern in Table C.16.

A remaining concern is that the rest-of-the-world benchmark is dominated by high-income economies in Europe and North America, whose industrial structures differ fundamentally from China's. If global robot demand shifted for reasons unrelated to labor supply, comparing China with advanced economies could overstate the China-specific surge. Table C.18 therefore re-estimates the aggregate specification against control groups of structurally more similar economies. Column (2) restricts the comparison to emerging manufacturing Asia (Vietnam, India, Thailand, Malaysia, and Indonesia); column (3) adds the major emerging manufacturing exporters outside Asia (Mexico, Brazil, and Türkiye); column (4) uses the East Asian manufacturing hubs (Japan, the Republic of Korea, Chinese Taipei, and Singapore); and column (5) uses a data-driven group, the ten economies whose pre-pandemic installation growth is closest to China's.

The differential acceleration survives every comparison. Relative to emerging manufacturing Asia, the group containing Vietnam, China's post-COVID installations rise by 0.61 log points more (column 2), with point estimates of 0.69 to 0.72 for the remaining groups; the East Asia hubs estimate is similar in magnitude. The corresponding stock estimates (Table C.19) are positive throughout and statistically significant in every comparison. The point estimates are naturally smaller than against the full country set, since part of the global gap reflects stagnating installations in advanced economies, but the central fact is that no structurally similar manufacturing exporter exhibits a surge comparable to China's.

## A.2 Policy Uncertainty and Automation: A Lockdown-Frequency Bartik IV

The baseline specification identifies the effect of the level of the migrant labor-supply shock. A distinct question is whether the unpredictability of that supply, not merely its decline, redirected innovation toward automation. When labor availability becomes unstable and politically contingent, firms have an incentive to automate as a hedge against operational risk, over and above the standard factor-price response (Bloom, 2009). To isolate this channel, we build a shift-share measure of imported policy uncertainty that parallels our baseline Bartik IV (Section 5.6) but replaces the mobility shift with the frequency of lockdowns in a migrant’s region.

**Lockdown-event data** We assemble a city-level lockdown-event dataset covering all 366 prefecture cities over 2020–2024, extending the keyword-based announcement-scraping method of Chen et al. (2025) from their 34-city sample to the full country.<sup>28</sup>

**Construction** Holding the 2015 migration-network shares  $s_{op}$  fixed (Equation 4), the imported policy-uncertainty exposure of destination county  $p$  in year  $t$  is

$$Z_{pt}^{\text{unc}} = \sum_o s_{op} n_{ot}. \quad (11)$$

where  $n_{ot}$  is the raw count of hard-lockdown events in origin city  $o$  in year  $t$ . A county draws high  $Z_{pt}^{\text{unc}}$  when the regions that historically supplied its migrants faced frequent, unpredictable lockdowns. Since a destination county does not choose how often its feeder regions lock down, this variation is plausibly exogenous to its own automation trajectory. To ensure  $Z_{pt}^{\text{unc}}$  captures imported rather than local disruption, we control for the county’s own prefecture lockdown frequency,  $\text{OwnFreq}_{pt} = n_{c(p),t}$ .

**Results** Table C.8 reports the estimates. The coefficient on  $Z_{pt}^{\text{unc}}$  is positive and significant for robot patent share: counties whose feeder regions faced more frequent, unpredictable lockdowns saw larger increases in automation patenting. The same exposure has no effect on total or non-automation patents, mirroring the baseline placebo pattern: uncertainty redirects innovation toward labor-saving technology specifically, not inventive activity in general. Because  $Z_{pt}^{\text{unc}}$  enters net of the county’s own lockdown and of the migrant-supply level captured by the baseline design, this result indicates that unpredictability is a distinct

<sup>28</sup>Because official notices rarely use the term *fengcheng* (city lockdown), we identify hard (citywide or main-urban) lockdowns from recurring administrative phrase: closed-off management of all areas.

driver of automation, beyond the level decline in migrant supply and its associated wage response.

### A.3 Heterogeneity in Migration Effects

**Heterogeneity by *hukou*** China's *hukou* system classifies individuals into agricultural and non-agricultural registration types, with important implications for labor mobility. Agricultural *hukou* holders face more institutional and economic barriers when working outside their hometowns, including limited access to urban social services and weaker integration into non-local labor markets. These constraints make them more vulnerable to disruptions during periods of restricted mobility.

Table C.10 presents the heterogeneity results by *hukou* type. The baseline interaction term—capturing the effect for agricultural *hukou* workers—is large, negative, and highly significant, indicating a sharp decline in out-of-hometown work for this group during the pandemic. In contrast, the triple-interaction coefficients for non-agricultural *hukou* holders are positive and significant across all specifications. These effects substantially offset the negative baseline impact.

**Heterogeneity by Gender** We also examine whether the impact of COVID-19 on cross-regional labor mobility differs by gender. In our specification, the baseline group consists of male workers, and the key interaction term captures the effect of COVID-19 on out-of-hometown work for pre-pandemic non-agricultural male workers. As reported in Table C.12, this coefficient is negative and highly significant across all columns (around  $-0.08$ ), indicating a sizable decline in migration among men employed in non-agricultural jobs before the pandemic.

The triple-interaction terms for women (Female  $\times$  Non-Agri  $\times$  Post-COVID) are also negative and statistically significant, with magnitudes of roughly  $-0.03$ . This implies that the overall reduction in out-of-hometown work is even larger for female workers than for their male counterparts. In other words, conditional on working in non-agricultural sectors prior to COVID-19, women experienced a stronger contraction in migration in the post-pandemic period. This pattern is consistent with the idea that women face additional constraints during crises—such as increased caregiving responsibilities and a higher concentration in service jobs that are more exposed to lockdowns and social distancing measures—which amplify the impact of mobility restrictions on their labor supply.

**Heterogeneity by Age** To examine life-cycle differences in the impact of COVID-19, individuals are grouped into three age categories: Age Group 1 (ages 35–49), Age Group 2



(ages 16–34), and Age Group 3 (ages 50–60). The prime-age group (35–49) serves as the reference category in the heterogeneity analysis.

Table C.13 presents the results. For the base group (Age 35–49), the interaction between being a pre-pandemic non-agricultural worker and the post-COVID period is negative and highly significant across all specifications (−0.096). This confirms a substantial decline in out-of-hometown work among prime-age migrants following the onset of COVID-19.

The triple-interaction terms reveal clear differences across age groups. For younger workers (Age 16–34), the coefficients are positive but statistically insignificant. This suggests that their reduction in migration was smaller than that of prime-age workers, but the difference is not large enough to be precisely estimated. In contrast, older workers (Age 50–60) show consistently negative and significant triple interactions across all columns, with magnitudes between −0.035 and −0.043. These results imply that older workers experienced a sharper contraction in cross-regional mobility relative to the prime-age reference group.

The results point to a clear life-cycle pattern: older workers reduced their migration the most, prime-age workers experienced substantial but moderate declines, and younger workers were the least affected. This gradient is consistent with heightened health risks and mobility constraints faced by older individuals during the pandemic.

#### A.4 Supply Side Consequences: Worker Reallocation, Informality, and Income

Having established that COVID-19 mobility frictions accelerated automation, we turn to the incidence of the shock on workers. To examine how displaced workers adjusted, via informality, unemployment, sectoral reallocation, and lost income, we estimate the following specification:

$$Y_{it} = \alpha + \beta(\text{MigrantWorker}_{i,\text{pre-COVID}} \times \text{COVID-19}_t) + \gamma_i + \delta_t + \phi_{pt} + \varepsilon_{it} \quad (12)$$

where  $Y_{it}$  is the outcome for individual  $i$  in year  $t$ . The treatment group comprises pre-pandemic migrant workers, compared against a control group of non-migrants. We control for individual fixed effects and province-year fixed effects.

Unlike our mobility analysis, which relied on non-agricultural employment as an intent-to-treat variable to avoid mechanical mean reversion, we define treatment here strictly by baseline migrant status. Because the dependent variables in this section are distinct from the treatment indicator, mechanical bounds are not a concern. More importantly, this tighter definition isolates the specific population exposed to the spatial lock-out. Using the broader non-agricultural category for worker-level outcomes would introduce severe

attenuation bias by confounding locked-out rural migrants with local urban workers, who faced no spatial barriers and potentially benefited from localized labor shortages.

Table C.22 reports the results. Columns (1) and (2) show that COVID-19 significantly raised the probability of informal employment among pre-pandemic migrant workers by 4.1–4.3 percentage points. This suggests that workers disrupted by migration restrictions did not smoothly transition into unemployment or alternative formal employment, but were instead pushed into the informal sector. This pattern is consistent with a well-established finding in the literature: the informal sector serves as a residual absorber of workers displaced by shocks to labor markets, technological change, and aggregate disruptions (Dix-Carneiro and Kovak, 2019; Ponczek and Ulyssea, 2022).

Turning to unemployment, columns (3) and (4) reveal a modest but statistically significant increase of 1.0–1.3 percentage points. While the magnitude is small, the result is consistent with the informality findings: workers displaced from formal employment largely sought income-generating activity in the informal sector rather than registering as unemployed. The sharp rise in informality thus accounts for the muted unemployment response, as the informal sector absorbed the bulk of displaced workers.

Columns (5) and (6) document the income consequences of the migration shock. Across specifications, the interaction between pre-pandemic migrant worker status and the post-2020 COVID-19 indicator is negative and statistically significant, implying that pre-pandemic migrant workers experienced a decline in annual income of approximately 8–9 percent following the outbreak of COVID-19 relative to workers who did not engage in cross-regional migration prior to the pandemic. These earnings losses are consistent with the disruption of the spatial matching process that underpins migrant labor markets: when migration is curtailed, workers are forced into lower-productivity local employment, depressing wages.

Finally, columns (7) and (8) examine sectoral reallocation out of manufacturing, restricting the sample to workers employed in manufacturing prior to the pandemic. The results suggest that the reallocation out of manufacturing was not concentrated among pre-pandemic migrant workers relative to non-migrants. This pattern is consistent with a broad-based disruption to manufacturing employment during the pandemic rather than one specifically mediated by migration status.

Figure B.9 plots the dynamic treatment effects by year and reveals a pattern: the effects persist and strengthen into 2022, well after mobility restrictions were lifted. If income losses and informality were purely mechanical consequences of being locked out, we would expect attenuation as restrictions eased. Instead, the persistence of effects beyond the restriction period suggests a more structural labor market deterioration, consistent with returning migrants encountering a more automated production environment, and



points to automation as a meaningful contributor to the distributional costs documented here.

Taken together, these findings show a picture of labor market adjustment in the wake of the pandemic-induced migration shock. Migrant workers faced meaningful income losses and were disproportionately pushed into informal employment, while unemployment rose modestly. The earnings consequences were sustained through the informal absorption of displaced workers rather than open unemployment, reflecting the income-smoothing role of informal work in the absence of adequate social insurance (Dix-Carneiro and Kovak, 2019).

## A.5 Back-of-the-Envelope: Aggregate Income Losses

The estimates of income losses among displaced migrant workers, rising informality, and firm-level investment responses can be aggregated to provide a rough sense of the macroeconomic stakes. We stress that the following calculations are illustrative: they combine our regression estimates with aggregate statistics under strong assumptions, and should be interpreted as order-of-magnitude benchmarks rather than precise accounting decompositions.

### A.5.1 Income Losses Among Displaced Migrant Workers

China's 2020 census counts approximately 376 million internal migrants. Of these, roughly 170 million were employed in manufacturing-intensive, migrant-dependent counties that constitute our treatment group. Our estimates in columns (5)–(6) of Table C.22 imply that pre-pandemic migrant workers in these counties experienced an 8–9 percent decline in annual income following COVID-19. Taking the midpoint (8.5 percent) and applying it to average migrant annual earnings of approximately 52,000 yuan in 2020 National Bureau of Statistics of China (2021), the per-worker income loss amounts to roughly 4,420 yuan per year.<sup>29</sup>

Multiplying across the affected migrant population:

$$\underbrace{170 \text{ million workers}}_{\text{treated migrants}} \times \underbrace{4,420 \text{ yuan}}_{\text{per-worker loss}} \approx 751 \text{ billion yuan} \approx \text{USD } 105 \text{ billion}. \quad (13)$$

At China's 2020 nominal GDP of approximately 101.6 trillion yuan, this represents roughly 0.74 percent of GDP, an aggregate income loss concentrated entirely among workers at the lower end of the income distribution.

<sup>29</sup>The number of 4,420 yuan is similar to the point estimate of migrant worker income loss in levels (e.g., 4,540 yuan per year).

### A.5.2 Informality Wage Penalty

The income loss estimate above captures the total earnings decline, which conflates the direct disruption to spatial matching with a composition effect: workers pushed into informality typically earn substantially less than their formal-sector counterparts. To isolate the informality channel, we apply a standard informal-sector wage penalty of 15–20 percent, consistent with estimates in the broader development literature (Dix-Carneiro and Kovak, 2019). Our estimates in columns (1)–(2) of Table C.22 indicate that pre-pandemic migrant workers experienced a 4.1–4.3 percentage point increase in informal employment probability. Applied to 170 million affected workers, this implies approximately 7 million additional workers displaced into informality. At an average formal-sector wage of 52,000 yuan and a 17.5 percent informal penalty, each newly informal worker faces an annual earnings shortfall of around 9,100 yuan, implying an informality-channel aggregate loss of approximately:

$$7 \text{ million workers} \times 9,100 \text{ yuan} \approx 64 \text{ billion yuan} \approx \text{USD } 9 \text{ billion}, \quad (14)$$

or roughly 0.06 percent of GDP. This is a partial lower bound: it captures only the wage penalty from formal-to-informal transitions, not the broader costs of reduced job security, foregone benefits, or occupational downgrading.

### A.5.3 Capital Reallocation on the Firm Side

Our estimates in Table C.6 imply that a one standard deviation increase in migrant dependence (1.23 percentage points) was associated with an 11,730 yuan increase in fixed assets per employee. China’s listed manufacturing firms employed approximately 12 million workers in the counties covered by our CSMAR sample. Scaling up, the aggregate increase in fixed asset investment attributable to the automation response amounts to roughly:

$$12 \text{ million employees} \times 11,730 \text{ yuan} \approx 141 \text{ billion yuan} \approx 0.14 \text{ percent of GDP}. \quad (15)$$

This capital reallocation represents a directed technical change response that substituted capital for labor (Acemoglu and Restrepo, 2018). While this investment may raise long-run productivity, in the short run it crowds out employment and depresses wages among low-skill workers, consistent with the labor market evidence above.

#### A.5.4 Total Labor Market Costs

Table C.23 consolidates these estimates. The largest component is the aggregate income loss among displaced migrant workers (approximately 0.74 percent of GDP), with an additional contribution from the informality wage penalty (0.06 percent). Taken together, the aggregate labor market disruption attributable to the COVID-induced automation episode amounts to approximately 0.80 percent of China’s 2020 GDP, a figure comparable in magnitude to estimates of the distributional costs of robot adoption in the United States (Acemoglu and Restrepo, 2020) and the labor market dislocations associated with “China Shock” (Autor et al., 2013). For reference, the memo row reports the aggregate increase in fixed asset investment among listed manufacturing firms (141 billion yuan, 0.14 percent of GDP), which represents the firm-side counterpart to the worker-side losses. Whether this capital deepening generates productivity gains sufficient to offset the distributional costs depends on the long-run returns to automation, which we cannot estimate from our data.

Three caveats bear emphasis. First, these estimates capture short-run transition costs and do not net out any long-run productivity gains from automation. Second, our CSMAR investment estimates cover only listed firms; if smaller unlisted firms, which employ the bulk of migrants, responded similarly, the capital reallocation figure would be substantially larger. Third, and most importantly, the income loss estimates capture the *total* labor market costs of the pandemic-induced automation episode rather than isolating the automation channel specifically. The observed income losses and informality transitions reflect a combination of two distinct mechanisms: the immediate disruption to spatial matching caused by mobility restrictions, which prevented workers from reaching their jobs, and the longer-run displacement effects of the subsequent automation investment, which rendered some of those jobs permanently obsolete. While our event study evidence suggests that the automation response is durable and persistent well beyond the mobility restriction period, implying that returning migrants would face a more automated production environment, we cannot cleanly decompose these costs between the two channels with the available data. The estimates should therefore be interpreted as upper bounds on the total labor market costs of the pandemic-induced automation episode rather than as precise measures of automation-attributable displacement, and we encourage future work with comprehensive employer-employee matched data to refine these figures.

### A.6 Additional Robustness Checks Using COVID-19 Cases

**Migrants and COVID-19 Cases** To assess the sensitivity of the baseline estimates to the definition of the COVID-19 shock, we replace the post-pandemic indicator with a continuous exposure measure constructed from prefecture-level confirmed COVID-19

cases in 2020 and 2022. Specifically, we use the logarithm of one plus the number of confirmed infections in each prefecture-year as an alternative measure of pandemic intensity. The results are reported in Table C.14. Across all specifications, the interaction between pre-pandemic non-agricultural employment and COVID-19 intensity remains negative, highly significant, and similar in magnitude, with coefficients ranging from  $-0.013$  to  $-0.019$ . These estimates indicate that the observed decline in cross-regional labor mobility is not driven by the binary timing structure of the baseline DID design but is robust to a more continuous measure of local pandemic severity.

**Alternative Measure: COVID-19 Intensity for Patents** If the mechanism driving automation is labor supply disruption, the effect should scale with pandemic severity. Regions with higher infection rates faced stricter lockdowns and more acute labor shortages. To test this, we replace the binary post-COVID indicator with a continuous exposure measure, interacting the pre-pandemic migrant share with the log number of confirmed COVID-19 cases at the prefecture level. Table C.15 reports the results. The interaction term is positive and statistically significant across all automation patent outcomes: a one unit increase in log COVID cases is associated with a 0.009 percentage point increase in robot patent share (column 1), a 0.8% increase in log robot counts (column 2), and a 2.8% increase in robots per capita (column 3). Consistent with the baseline placebo tests, the interaction is small and insignificant for total and non-automation patents (columns 5 and 6). This dose-response pattern confirms that the automation response scales with pandemic intensity and is specifically directed toward labor-saving technologies rather than reflecting a general increase in innovative activity.

## A.7 Sensitivity to Omitted Variable Bias: A Diegert et al. Analysis

A natural concern is that the estimated relationship between migrant COVID-19 exposure and automation reflects an omitted regional characteristic correlated with both migrant outflows and automation incentives, rather than the labor supply shock itself. A leading candidate is that the pandemic coincided with a substantial expansion of government R&D subsidies and firm-level innovation spending in China (Mao, 2024), which could raise automation independently of local migrant exposure. China’s domestic technology policy in this period was explicitly oriented toward indigenous innovation in advanced manufacturing (Naughton, 2021). If this policy push intensified during the pandemic and happened to coincide with the geography of migrant-intensive counties, it could generate a spurious positive association in our baseline specification.

We assess this concern using the sensitivity analysis framework of Diegert et al. (2022),

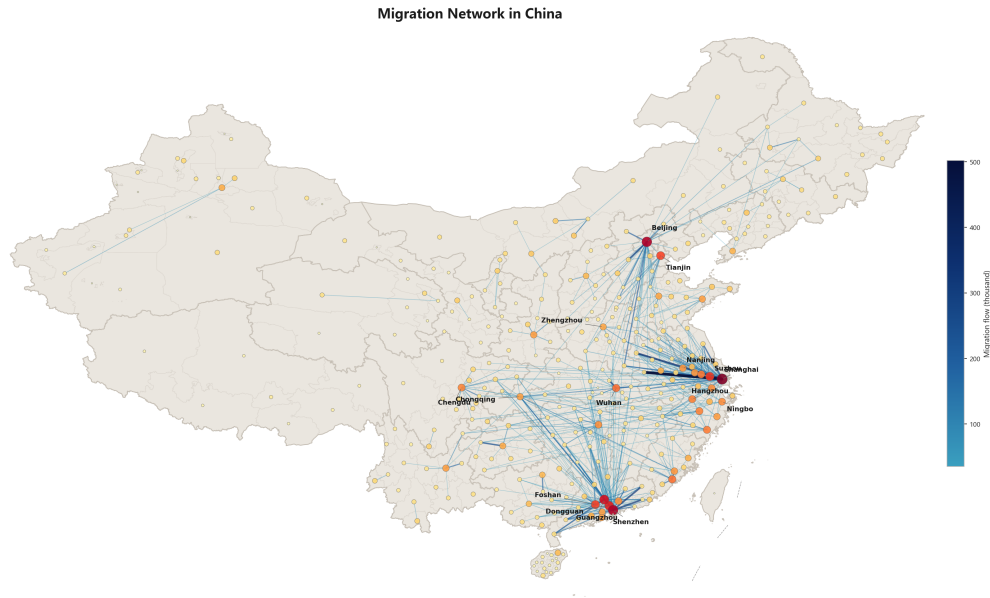
hereafter DMP, which traces the identified set for the treatment effect as a function of two sensitivity parameters:  $\bar{r}_x$ , the strength of an unobserved confounder's association with the treatment relative to an observed calibration benchmark, and  $\bar{c}$ , the degree to which the confounder may be correlated with the included calibration covariates. The framework does not require the benchmark to be exogenous to the confounder, which matters here because the R&D channel is itself plausibly endogenous.

We calibrate the analysis against the R&D channel directly. Figure B.10 plots the resulting identified sets for  $\bar{c} \in \{0, 0.5, 1\}$ , and Table C.21 reports the corresponding breakdown points. Panel (a) (Figure B.10a) uses county (log) patent stock as the calibration benchmark; Panel (b) (Figure B.10b) adds prefecture science-and-technology expenditure; and Panel (c) (Figure B.10c) adds prefecture-level listed-firm R&D spending. In every panel the bound curves converge, at low  $\bar{r}_x$ , on a small positive treatment effect, and the point estimate is stable in sign and magnitude across the three calibrations. The R&D covariates therefore do not mechanically absorb the migrant-exposure effect.

As expected, each identified set widens as the assumed confounder grows stronger ( $\bar{r}_x$ ) and more correlated with the calibration covariates ( $\bar{c}$ ). Under the patent anchor (Panel a) and the listed-firm R&D specification (Panel c), the lower bound remains above zero across most of the plotted range: breakdown occurs only beyond  $\bar{r}_x \approx 1$  for  $\bar{c} = 0.5$  and  $\bar{c} = 1$ , while the  $\bar{c} = 0$  bound stays positive almost to  $\bar{r}_x = 2$ . Under the science-and-technology specification (Panel b) the bounds widen more rapidly and breakdown points fall closer to  $\bar{r}_x \approx 0.5$ . We interpret these lower values with care: as DMP note, a more strongly predictive calibration covariate mechanically reduces the relative breakdown point even when the point estimate is unaffected, so the stability of  $\hat{\beta}$  across calibrations is the more informative indicator of robustness. Taken together, the results indicate that a confounder substantially stronger than the documented R&D expansion would be required to overturn the central finding.

## B Figures

Figure B.1: Migration Networks in China



*Notes:* This figure displays the bilateral migration networks across Chinese cities, constructed from the 2015 China 1% Population Survey (mini-census). Each node represents a city and each line represents a migration flow between origin and destination cities. The thickness of the lines reflects the intensity of bilateral migration flows. Data are aggregated from individual-level records by observing both the registered *hukou* location (origin) and current place of residence (destination) of each surveyed individual.

Figure B.2: Enforcement of Digital Health Codes for Inter-Regional Travel



*Notes:* This figure illustrates a typical digital checkpoint at a railway transit hub during the post-lockdown pandemic period. The blue sign requires passengers to present a valid “green” code via the local municipal health application (in this case, the Beijing Health Kit) before being permitted to board. These widespread digital checkpoints served as binding constraints on the cross-regional mobility of migrant workers.

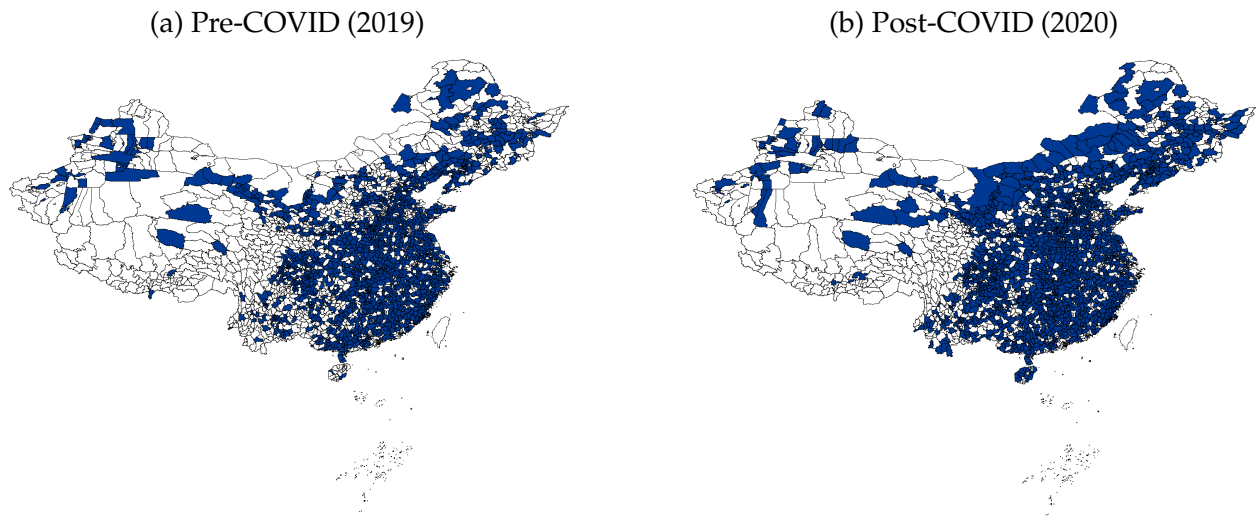
Figure B.3: The Three-Tier Color-Coded Health Classification System



*Notes:* This figure displays a representative municipal health code interface (the Hangzhou Health Code), which dynamically classified individuals based on nucleic acid test results and travel history. A Green code permitted normal mobility, a Yellow code mandated a 7-day quarantine, and a Red code mandated a 14-day quarantine. For migrant workers, losing green code status effectively halted their ability to travel for work or access urban labor markets, creating severe, high-frequency labor supply shocks for manufacturing and construction firms.



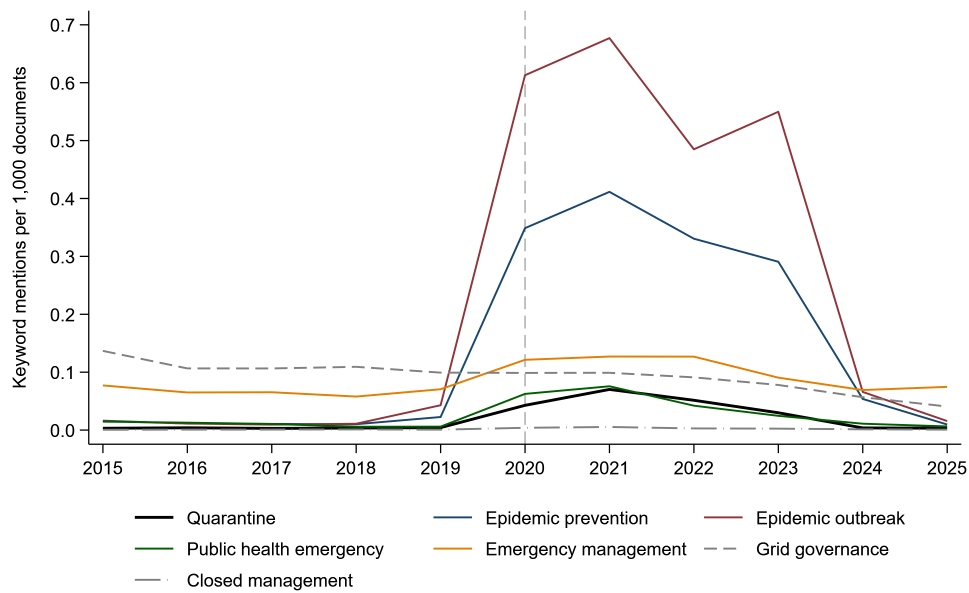
Figure B.4: Robot Patent Activity Before and After COVID-19



*Notes:* This figure displays the extensive margin of robot patent activity across Chinese counties before and after the COVID-19 shock. Panel (a) shows counties with at least one robot-related patent application in 2019, and Panel (b) shows counties with at least one robot-related patent application after 2020. A county is defined as having robot patent activity if it filed at least one patent related to robots, manipulators, or automated guided vehicles (AGVs) in the given year. Data are drawn from the China National Intellectual Property Administration (CNIPA).

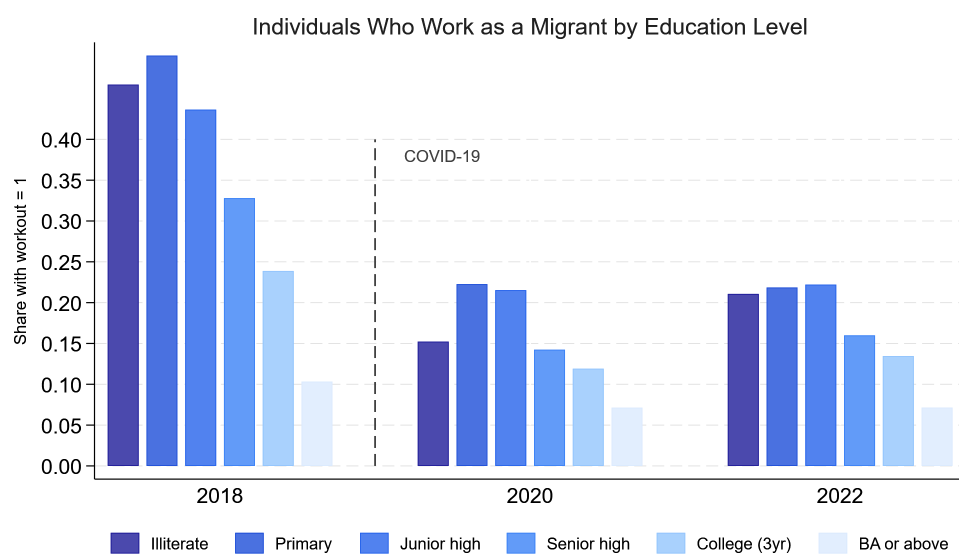


Figure B.5: COVID-Related Keyword Trends in Chinese Local Government Policy Documents



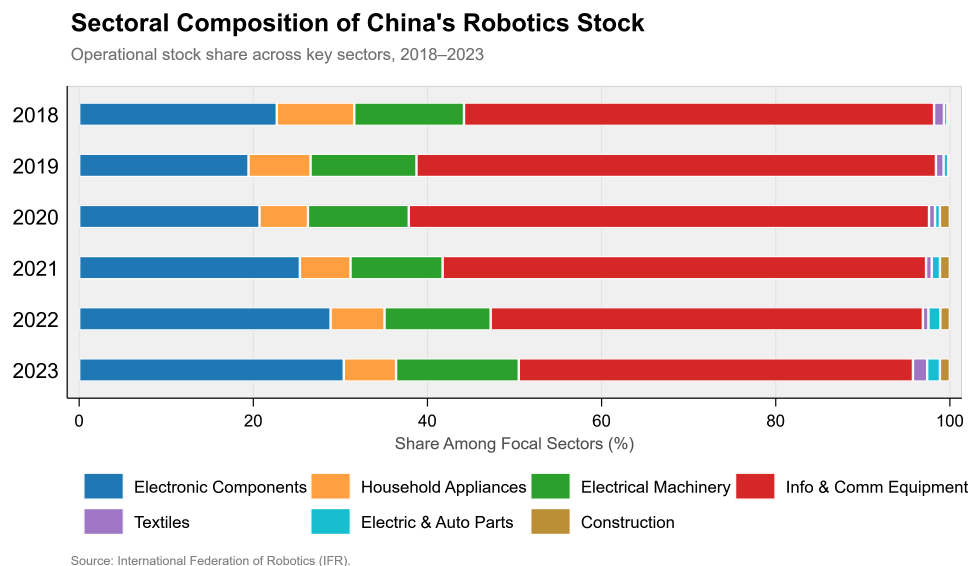
*Notes:* Each series shows the average number of keyword mentions per 1,000 words across county-level government policy documents. The dashed vertical line marks 2020, the onset of the COVID-19 pandemic. Keywords are extracted from Chinese local government work reports.

Figure B.6: Presence of Migrant Workers by Education Level



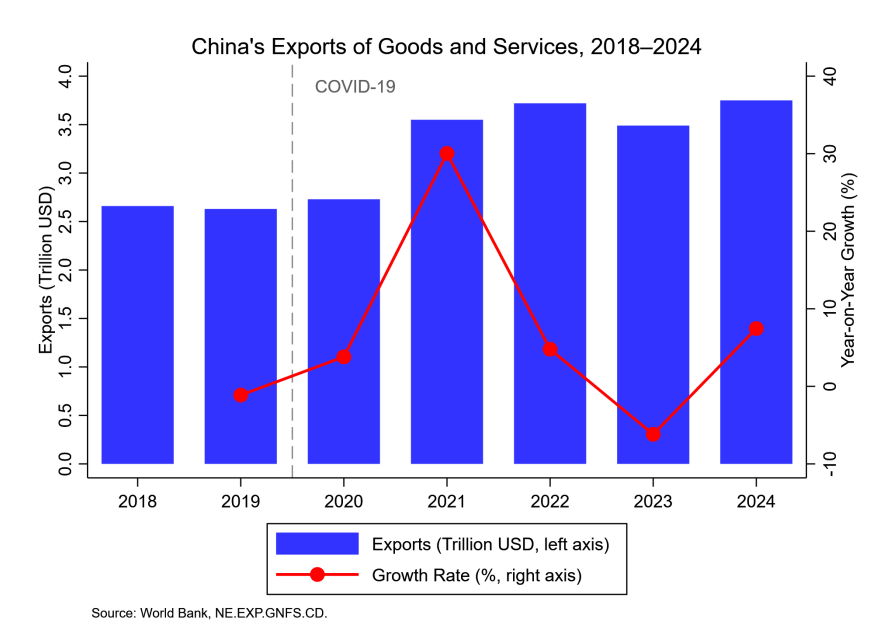
*Notes:* This figure displays the share of migrant workers by education level using CFPS data. Education categories range from no schooling to college and above.

Figure B.7: Decomposition of Robotics by Industries in China Before and After COVID-19



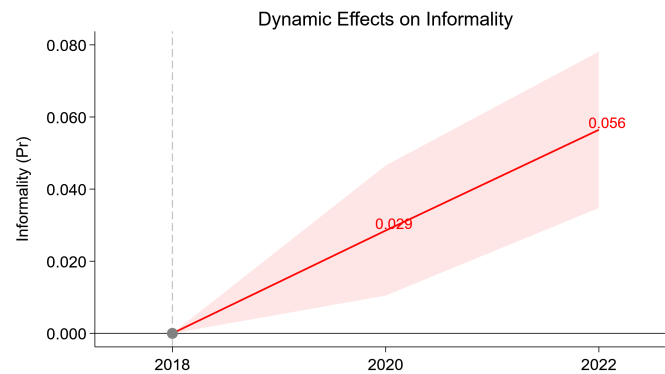
*Notes:* This figure displays the sectoral composition of industrial robot stock in China, comparing the pre-COVID and post-COVID periods. Each bar shows the share of total robot stock attributed to each industry. Data are drawn from the International Federation of Robotics (IFR).

Figure B.8: China's Exports of Goods and Services, 2018–2024

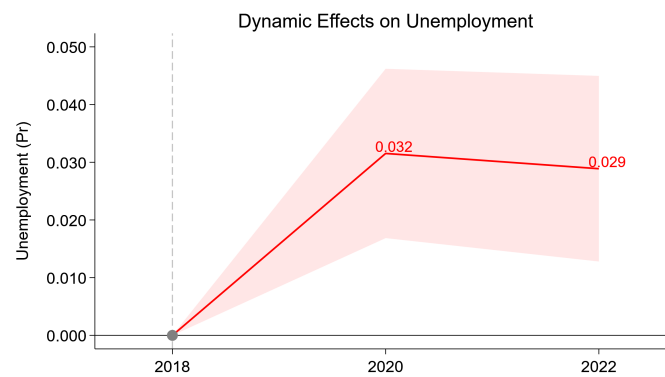


*Notes:* This figure plots China's total exports of goods and services in trillion USD (bars, left axis) and the year-on-year growth rate (line, right axis) from 2018 to 2024. The dashed vertical line marks the onset of the COVID-19 pandemic in 2020. Data are from the World Bank National Accounts.

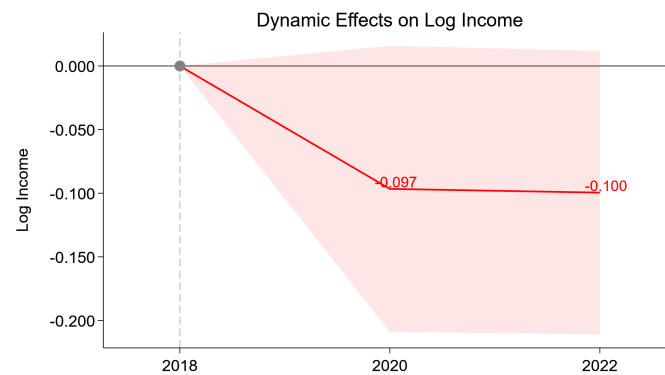
Figure B.9: Dynamic Effects of Migrant-Work Exposure on Labor Market Outcomes



(a) Informality



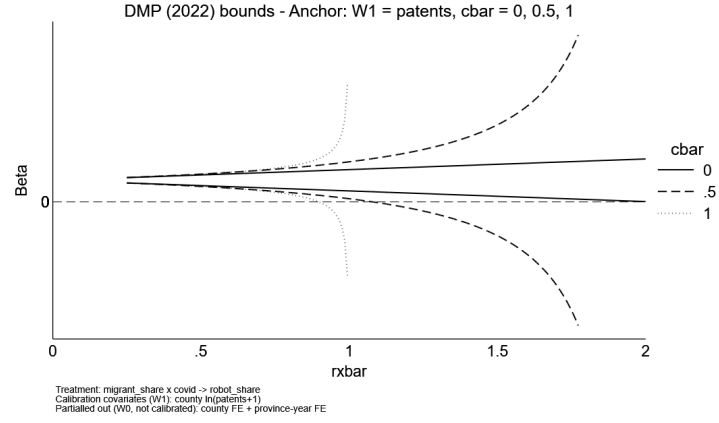
(b) Unemployment



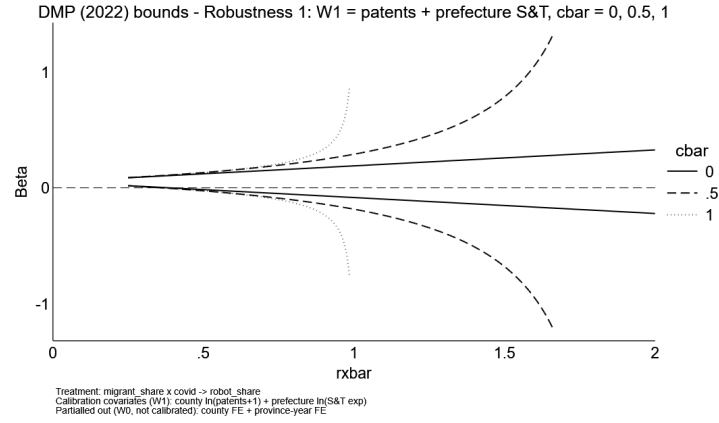
(c) Log Income

*Notes:* Each panel plots the coefficients from interacting an indicator for working outside one's hometown in 2018 with year dummies, relative to the 2018 baseline (normalized to zero). The sample covers CFPS waves 2018, 2020, and 2022. All regressions include individual fixed effects and province  $\times$  year fixed effects. Standard errors are clustered at the individual level. The 95% confidence intervals are shown as vertical bars.

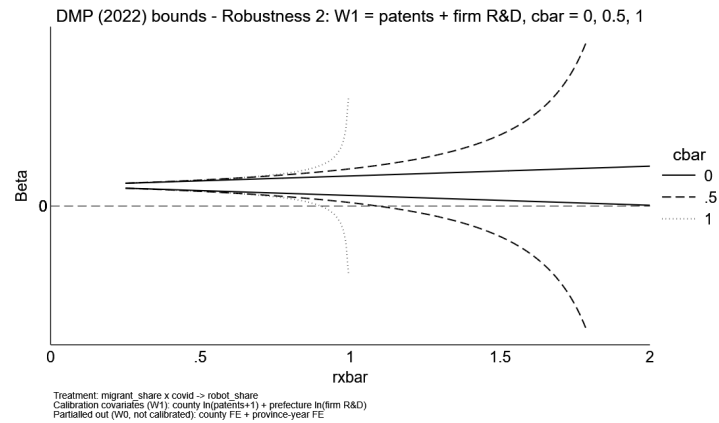
Figure B.10: [Diegert et al. \(2022\)](#) Identified Set Under R&D-Channel Calibrations



(a) Anchor:  $W_1 = \text{patents}$



(b)  $W_1 = \text{patents} + \text{prefecture S\&T}$



(c)  $W_1 = \text{patents} + \text{listed-firm R\&D}$

Notes: This figure plots the identified set for the coefficient on migrant exposure interacted with the post-COVID-19 indicator, following [Diegert et al. \(2022\)](#), as a function of the sensitivity parameter  $rxbar$ , across  $\bar{c} \in \{0, 0.5, 1\}$ . All panels partial out county and province-by-year fixed effects ( $W_0$ ). Calibration covariates ( $W_1$ ) are residualized on  $W_0$  and differ by panel: Panel (a) uses county log patent stock as an anchor; Panel (b) adds prefecture science and technology expenditure; Panel (c) adds listed-firm R&D spending. The horizontal dashed line denotes zero; the breakdown point is the value of  $rxbar$  at which the identified set first includes zero.

# C Tables

Table C.1: Sample Automation Patent from the CNIPA Database

| Example Patent Record: Automation Technology |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title (Original)                             | 一种汽车零部件 自动化焊接用机器人                                                                                                                                                                                                                                                                                                                                                                                                            |
| Title (Translated)                           | A <b>Robot</b> for <b>Automated</b> Welding of Automotive Parts                                                                                                                                                                                                                                                                                                                                                              |
| Patent ID                                    | CN202120476704.1                                                                                                                                                                                                                                                                                                                                                                                                             |
| Application Date                             | 2021-03-05 (Post-COVID)                                                                                                                                                                                                                                                                                                                                                                                                      |
| Applicant                                    | Shanghai Aocen Automation Technology Co., Ltd.                                                                                                                                                                                                                                                                                                                                                                               |
| Location                                     | Jiading District, Shanghai                                                                                                                                                                                                                                                                                                                                                                                                   |
| IPC Code                                     | B23K37/00 (Auxiliary devices for welding)                                                                                                                                                                                                                                                                                                                                                                                    |
| Abstract (Translated)                        | The utility model discloses a <b>robot for automated welding</b> of automotive parts. It belongs to the technical field of welding robots. The device includes a base, a rotating arm, a fan, and an exhaust system. Through the coordination of the suction hood and pipes, it effectively diverts gas generated during the welding process, improving visibility for the operator and reducing environmental interference. |

*Note:* This patent illustrates a labor-saving technology (welding robot) filed in a manufacturing hub (Shanghai) after the pandemic outbreak. Our text analysis algorithm identifies this as an automation patent based on the keywords “Robot”.

Table C.2: Sample Manipulator Patent from the CNIPA Database

| Example Patent Record: Automation Technology |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title (Original)</b>                      | 一种加工定位方法及柔性工装系统                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Title (Translated)</b>                    | A Processing Positioning Method and <b>Flexible</b> Tooling System                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Patent ID</b>                             | CN202110165235.6                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Application Date</b>                      | 2021-02-06 (Post-COVID)                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Applicant</b>                             | Shanghai University                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Location</b>                              | Baoshan District, Shanghai                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>IPC Code</b>                              | B23Q3/00 (Devices for holding workpieces)                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Abstract (Translated)</b>                 | The invention discloses a positioning method and flexible tooling system. By utilizing a <b>robotic arm</b> (mechanical arm) to drive a six-degree-of-freedom platform, the system significantly improves automation. It is designed for processing workpieces with similar structures but varying positioning requirements, effectively reducing tooling design time and production cycles while enhancing positioning accuracy and efficiency. |

*Note:* This patent illustrates the adoption of “flexible” manufacturing technology filed in Shanghai immediately following the first pandemic wave. Our text analysis identifies this as a relevant outcome based on the keywords “Robotic Arm” in the abstract.



Table C.3: Descriptive Statistics: Individual Characteristics and Intent to Treatment (Migration)

|                        | Full Sample |        | Treatment = 1<br>(Non-agri Worker) |        | Treatment = 0<br>(Agricultural Worker) |        |
|------------------------|-------------|--------|------------------------------------|--------|----------------------------------------|--------|
|                        | Mean        | SD     | Mean                               | SD     | Mean                                   | SD     |
| Migrant worker         | 0.181       | 0.385  | 0.251                              | 0.433  | 0.114                                  | 0.318  |
| Female                 | 0.438       | 0.496  | 0.382                              | 0.486  | 0.492                                  | 0.500  |
| Age                    | 46.640      | 13.640 | 41.138                             | 11.610 | 51.882                                 | 13.363 |
| Non-agric <i>hukou</i> | 0.221       | 0.415  | 0.353                              | 0.478  | 0.097                                  | 0.296  |
| Education level        | 2.885       | 1.425  | 3.509                              | 1.370  | 2.290                                  | 1.205  |
| N                      | 37,377      |        | 18,237                             |        | 19,140                                 |        |

*Notes:* This table reports summary statistics for individual characteristics from the China Family Panel Studies (CFPS). Treatment is defined as individuals engaged in non-agricultural work prior to the pandemic (Treat = 1) versus those in agricultural work (Treat = 0). Education level is measured on an eight-level scale.

Table C.4: City-Level Lockdown Events in China, 2020–2022

| Year                                                                                 | Total events | Partial lockdowns | Full lockdowns | Cities with any event | Cities with full lockdown | Mean events per city |
|--------------------------------------------------------------------------------------|--------------|-------------------|----------------|-----------------------|---------------------------|----------------------|
| <i>Panel A: Lockdown events by year (366 prefecture-level cities)</i>                |              |                   |                |                       |                           |                      |
| 2020                                                                                 | 414          | 275               | 139            | 183                   | 46                        | 1.13                 |
| 2021                                                                                 | 394          | 222               | 172            | 101                   | 37                        | 1.08                 |
| 2022                                                                                 | 965          | 659               | 306            | 228                   | 53                        | 2.64                 |
| <i>Panel B: Distribution of events per city, first three COVID years (2020–2022)</i> |              |                   |                |                       |                           |                      |
|                                                                                      | Mean         | Median            | SD             | P25 / P75             | P90                       | Max                  |
| Lockdown events per city                                                             | 4.84         | 3                 | 5.04           | 1 / 7                 | 13                        | 23                   |
| Total events: 1,773 (partial 1,156; full 617)                                        |              |                   |                |                       |                           |                      |

*Notes:* This table summarizes the city-level lockdown-event dataset constructed from keyword-based scraping of local government lockdown announcements (Baidu Search), following the method of [Chen et al. \(2025\)](#), for all 366 prefecture-level cities over 2020–2022. An event is a dated lockdown announcement; *full* denotes a citywide or main-urban-area hard lockdown, *partial* a district- or neighborhood-level lockdown. China ended formal lockdowns in December 2022; the small number of post-2022 detections reflects retrospective coverage and is set to zero in the shift-share analysis (Appendix [A.2](#)).

Table C.5: Counties Most and Least Exposed to Migrants

| Most Exposed Counties |               |                | Least Exposed Counties |                |                |
|-----------------------|---------------|----------------|------------------------|----------------|----------------|
| County                | City          | Province       | County                 | City           | Province       |
| Nanhai                | Foshan        | Guangdong      | Dongyuan               | Heyuan         | Guangdong      |
| Yiwu                  | Yiwu          | Zhejiang       | Ji                     | Linfen         | Shanxi         |
| Shunde                | Foshan        | Guangdong      | Zhuozi                 | Ulanqab        | Inner Mongolia |
| Ruili                 | Ruili         | Yunnan         | Liangzihu              | Ezhou          | Hubei          |
| Aksay Kazakh A.C.     | Jiuquan       | Gansu          | Pengan                 | Nanchong       | Sichuan        |
| Jimei                 | Xiamen        | Fujian         | Yingshang              | Fuyang         | Anhui          |
| Huiyang               | Huizhou       | Guangdong      | Huaiji                 | Zhaoqing       | Guangdong      |
| Chenggong             | Kunming       | Yunnan         | Taipusi Banner         | Xilin Gol      | Inner Mongolia |
| Shishi                | Shishi        | Fujian         | Qinghemmen             | Fuxin          | Liaoning       |
| Kunshan               | Kunshan       | Jiangsu        | Anju                   | Suining        | Sichuan        |
| Huolinguole           | Huolinguole   | Inner Mongolia | Guyang                 | Baotou         | Inner Mongolia |
| Erenhot               | Erenhot       | Inner Mongolia | Huoqiu                 | Lu'an          | Anhui          |
| Bayan Obo Mining      | Baotou        | Inner Mongolia | Qingjian               | Yulin          | Shaanxi        |
| Furong                | Changsha      | Hunan          | Yantan                 | Zigong         | Sichuan        |
| Suifenhe              | Suifenhe      | Heilongjiang   | Kangbao                | Zhangjiakou    | Hebei          |
| Hainan District       | Wuhai         | Inner Mongolia | Xinghai                | Hainan Tibetan | Qinghai        |
| Sanshui               | Foshan        | Guangdong      | Urumqi County          | Urumqi         | Xinjiang       |
| Haicang               | Xiamen        | Fujian         | Lixin                  | Bozhou         | Anhui          |
| Gangkou               | Fangchenggang | Guangxi        | Liangcheng             | Ulanqab        | Inner Mongolia |
| Simao                 | Pu'er         | Yunnan         | Shangdu                | Ulanqab        | Inner Mongolia |
| Gongshan D.N. A.C.    | Nujiang Lisu  | Yunnan         | Xiuyu                  | Putian         | Fujian         |
| Xishan                | Wuxi          | Jiangsu        | Zizhou                 | Yulin          | Shaanxi        |
| Dongsheng             | Ordos         | Inner Mongolia | Zhaohua                | Guangyuan      | Sichuan        |
| Wujin                 | Changzhou     | Jiangsu        | Anyi                   | Nanchang       | Jiangxi        |
| Beichen               | Tianjin       | Tianjin        | Fengzhen               | Fengzhen       | Inner Mongolia |

Notes – This table lists the top and bottom 25 counties by migrant share, defined as net migration divided by 2015 total population. A.C. = Autonomous County. D.N. = Dulong-Nu. Counties with missing population in 2015 are excluded.

Table C.6: Effect of COVID-19 on Upstream Automation Investment of Firms

| VARIABLES                         | ln(R&D<br>Spending+1)<br>(1) | ln(R&D<br>Personnel+1)<br>(2) | R&D Spending<br>Ratio<br>(3) | R&D Personnel<br>Ratio<br>(4) | Fixed Assets<br>per Employee<br>(5) |
|-----------------------------------|------------------------------|-------------------------------|------------------------------|-------------------------------|-------------------------------------|
| Migrant Share % $\times$ COVID-19 | 0.119**<br>(0.057)           | 0.069***<br>(0.020)           | 1.155<br>(0.883)             | 0.159**<br>(0.064)            | 9,541.632*<br>(5,524.748)           |
| Observations                      | 21,568                       | 21,568                        | 21,568                       | 21,568                        | 21,568                              |
| R-squared                         | 0.936                        | 0.937                         | 0.180                        | 0.857                         | 0.818                               |
| County FEs                        | ✓                            | ✓                             | ✓                            | ✓                             | ✓                                   |
| Province $\times$ Year FEs        | ✓                            | ✓                             | ✓                            | ✓                             | ✓                                   |
| Initial GDP $\times$ Year FEs     | ✓                            | ✓                             | ✓                            | ✓                             | ✓                                   |

*Notes:* This table reports the effect of pandemic-induced labor mobility restrictions on firm-level upstream automation investment, collapsed to the county level using listed firm data from CSMAR. The key independent variable is the interaction between county-level migrant share (2015) and COVID-19 onset. All specifications include county fixed effects and interactions of initial GDP per capita and migrant share with year fixed effects. Standard errors clustered at the county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.7: Alternative Explanation: Local Government Industrial Policy Priority and Migrant Labor Scarcity

| VARIABLES                         | Automation-related Keyword Frequency in County Government Work Reports (per 1,000 words) |                         |                             |                            |                           |                         |                       |
|-----------------------------------|------------------------------------------------------------------------------------------|-------------------------|-----------------------------|----------------------------|---------------------------|-------------------------|-----------------------|
|                                   | Robot Substitution<br>(1)                                                                | Industrial Robot<br>(2) | Automation Equipment<br>(3) | Smart Manufacturing<br>(4) | Industry Upgrading<br>(5) | Unmanned Factory<br>(6) | Total Keywords<br>(7) |
| Migrant Share % $\times$ COVID-19 | −0.000<br>(0.000)                                                                        | −0.001<br>(0.002)       | −0.000<br>(0.000)           | −0.002<br>(0.001)          | −0.000<br>(0.000)         | 0.000<br>(0.000)        | 0.001<br>(0.005)      |
| Observations                      | 15,159                                                                                   | 15,159                  | 15,159                      | 15,159                     | 15,159                    | 15,159                  | 15,159                |
| R-squared                         | 0.143                                                                                    | 0.249                   | 0.155                       | 0.216                      | 0.221                     | 0.170                   | 0.304                 |
| County FEs                        | ✓                                                                                        | ✓                       | ✓                           | ✓                          | ✓                         | ✓                       | ✓                     |
| Province $\times$ Year FEs        | ✓                                                                                        | ✓                       | ✓                           | ✓                          | ✓                         | ✓                       | ✓                     |
| Initial GDP $\times$ Year FEs     | ✓                                                                                        | ✓                       | ✓                           | ✓                          | ✓                         | ✓                       | ✓                     |

*Notes:* This table tests whether the observed automation response was driven by local government industrial policy rather than firm-level responses to labor scarcity. Dependent variables measure the frequency of automation-related keywords per 1,000 words in county-level government work reports, a standard proxy for local policy priority. The keyword categories are: robot substitution (Column 1), industrial robots (Column 2), automation equipment (Column 3), smart manufacturing (Column 4), industry upgrading (Column 5), unmanned factories (Column 6), and a composite total across all categories (Column 7). If counties facing greater migrant labor disruption simultaneously intensified local automation policy, the main estimates could reflect government-directed investment rather than decentralized firm-level responses. The uniformly small and statistically insignificant coefficients across all seven columns rule out this interpretation: counties with larger migrant shares did not experience systematically greater government emphasis on automation during the COVID-19 period. The automation response is therefore unlikely to be an artifact of local industrial policy. Migrant Share is net migration as a share of total population in 2015. COVID-19 indicates years 2020–2023. All specifications include county fixed effects, province  $\times$  year fixed effects, and interactions of initial GDP per capita with year fixed effects. Standard errors clustered at county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.8: Bartik Instrumental Variable: Origin and Destination Lockdown Frequency

| VARIABLES                                          | Robot Patent<br>Share %<br>(1) | ln(Total<br>Patent+1)<br>(2) | ln(Non-Auto<br>Patent+1)<br>(3) |
|----------------------------------------------------|--------------------------------|------------------------------|---------------------------------|
| $Z_{p,t}^{\text{freq}}$ (Origin Lockdown Exposure) | 0.153**<br>(0.069)             | 0.007<br>(0.019)             | 0.005<br>(0.019)                |
| Own Lockdown Frequency                             | -0.048<br>(0.043)              | 0.004<br>(0.011)             | 0.005<br>(0.011)                |
| Observations                                       | 10,985                         | 10,985                       | 10,985                          |
| R-squared                                          | 0.294                          | 0.966                        | 0.966                           |
| County FEs                                         | ✓                              | ✓                            | ✓                               |
| Province $\times$ Year FEs                         | ✓                              | ✓                            | ✓                               |
| Initial GDP $\times$ Year FEs                      | ✓                              | ✓                            | ✓                               |

*Notes:* This table reports Bartik IV estimates using an alternative shift measure based on lockdown frequency rather than the Baidu outflow index.  $Z_{p,t}^{\text{freq}}$  is the share-weighted sum of origin-city lockdown frequency, constructed analogously to  $Z_{p,t}$  in Table 5 by replacing the outflow shift  $g_{ot}$  with the frequency of lockdown measures announced in each origin city. Own Lockdown Frequency is the destination county's own lockdown frequency, included to isolate the effect of migrant labor supply disruption transmitted through the origin network from any direct, local effect of destination-county lockdown measures. Column (1) reports robot patent share; Columns (2) and (3) report placebo outcomes, total patents and non-automation patents, respectively. All specifications include county fixed effects and province  $\times$  year fixed effects. Standard errors clustered at the county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.9: Migrant-Sending Counties Experienced the Out-Migration Collapse

| VARIABLES                          | (1)<br>Out-of-hometown Work Rate |
|------------------------------------|----------------------------------|
| Origin Migrant Share $\times$ 2020 | −0.0400**<br>(0.0186)            |
| Origin Migrant Share $\times$ 2022 | −0.0248<br>(0.0210)              |
| Observations                       | 378                              |
| R-squared                          | 0.663                            |
| County FEs                         | ✓                                |
| Year FEs                           | ✓                                |

*Notes:* This table reports the county-level first stage of the labor supply shock. The dependent variable is the share of CFPS respondents in a county working outside their hometown, computed from individual responses collapsed to county-year cells (2018, 2020, and 2022 waves). Origin migrant share is the county's out-migration intensity from the 2015 mini-census (out-migrants over county population, winsorized at the 1st and 99th percentiles), interacted with year indicators; 2018 is the omitted base year, and the level effect is absorbed by the county fixed effects. Estimates are weighted by 2015 county population. A county with a 10 percentage point higher pre-pandemic out-migration share experienced a 0.4 percentage point larger decline in its out-of-hometown work rate by 2020; the 2022 interaction remains negative but is imprecisely estimated. Because CFPS samples households at their home location, this specification captures the collapse of out-migration from sending counties, which transmits to migrant-receiving destination counties through pre-existing migration networks, the variation exploited by the shift-share design in Appendix 5.6. Standard errors clustered at the county level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.10: Heterogeneity: The Effect of COVID-19 by *hukou*

| VARIABLES                                            | Out-of-Hometown Work |                      |                      |
|------------------------------------------------------|----------------------|----------------------|----------------------|
|                                                      | (1)                  | (2)                  | (3)                  |
| <i>Base Effect (Agricultural hukou)</i>              |                      |                      |                      |
| Non-Agri $\times$ COVID-19                           | -0.153***<br>(0.009) | -0.159***<br>(0.011) | -0.165***<br>(0.010) |
| <i>Interactions (Relative to Agricultural hukou)</i> |                      |                      |                      |
| $\times$ Non-agric <i>hukou</i>                      | 0.167***<br>(0.022)  | 0.154***<br>(0.025)  | 0.170***<br>(0.022)  |
| Observations                                         | 34,785               | 34,351               | 34,785               |
| R-squared                                            | 0.631                | 0.464                | 0.638                |
| Individual FEs                                       | ✓                    |                      | ✓                    |
| Household FEs                                        |                      | ✓                    |                      |
| Year FEs                                             | ✓                    | ✓                    |                      |
| Province $\times$ Year FEs                           |                      |                      | ✓                    |

*Notes:* This table reports heterogeneity in the COVID-19 effect by *hukou* status. The dependent variable is an indicator for out-of-hometown migrant work. The base category is agricultural *hukou* holders, so the coefficient on Non-Agri  $\times$  COVID-19 reflects the effect for this group. The interaction term captures the differential effect for non-agricultural *hukou* holders. Column (1) includes individual and year fixed effects; Column (2) includes household and year fixed effects; Column (3) includes individual and province-by-year fixed effects. Standard errors clustered at the individual level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .



Table C.11: Heterogeneity: The Effect of COVID-19 by Skill Level (Proxy by Education)

| VARIABLES                                                     | Out-of-Hometown Work |                      |                      |
|---------------------------------------------------------------|----------------------|----------------------|----------------------|
|                                                               | (1)                  | (2)                  | (3)                  |
| <i>Base Effect (Illiterate or semi-literate)</i>              |                      |                      |                      |
| Non-Agri $\times$ COVID-19                                    | -0.228***<br>(0.024) | -0.233***<br>(0.027) | -0.227***<br>(0.024) |
| <i>Interactions (Relative to Illiterate or semi-literate)</i> |                      |                      |                      |
| $\times$ Primary School                                       | 0.047<br>(0.030)     | 0.045<br>(0.033)     | 0.035<br>(0.030)     |
| $\times$ Middle School                                        | 0.147***<br>(0.028)  | 0.140***<br>(0.031)  | 0.134***<br>(0.028)  |
| $\times$ High School                                          | 0.191***<br>(0.032)  | 0.176***<br>(0.036)  | 0.176***<br>(0.032)  |
| $\times$ 3-Year College / Vocational                          | 0.219***<br>(0.054)  | 0.175***<br>(0.056)  | 0.220***<br>(0.054)  |
| $\times$ BA or above                                          | 0.260***<br>(0.043)  | 0.213***<br>(0.057)  | 0.250***<br>(0.043)  |
| Observations                                                  | 37,336               | 36,838               | 37,200               |
| R-squared                                                     | 0.640                | 0.469                | 0.644                |
| Individual FEs                                                | ✓                    |                      | ✓                    |
| Household FEs                                                 |                      | ✓                    |                      |
| Year FEs                                                      | ✓                    | ✓                    |                      |
| Province $\times$ Year FEs                                    |                      |                      | ✓                    |

*Notes:* The dependent variable is an indicator for out-of-hometown migrant work. The base education category is “Illiterate or semi-literate”, and the coefficient on Non-Agri  $\times$  COVID-19 therefore reflects the COVID-19 period effect for the lowest-education group. The interaction terms show how this effect differs for individuals with higher schooling levels. Column (1) includes individual and year fixed effects; Column (2) includes household and year fixed effects; Column (3) includes individual and province-by-year fixed effects. Standard errors clustered at the individual level are reported in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.12: Heterogeneity: The Effect of COVID-19 by Gender

| VARIABLES                              | Out-of-Hometown Work |                      |                      |
|----------------------------------------|----------------------|----------------------|----------------------|
|                                        | (1)                  | (2)                  | (3)                  |
| <i>Base Effect (Male)</i>              |                      |                      |                      |
| Non-Agri $\times$ COVID-19             | -0.075***<br>(0.010) | -0.091***<br>(0.012) | -0.087***<br>(0.011) |
| <i>Interactions (Relative to Male)</i> |                      |                      |                      |
| $\times$ Female                        | -0.029**<br>(0.014)  | -0.027*<br>(0.016)   | -0.034**<br>(0.014)  |
| Observations                           | 37,377               | 36,880               | 37,239               |
| R-squared                              | 0.637                | 0.468                | 0.642                |
| Individual FEs                         | ✓                    |                      | ✓                    |
| Household FEs                          |                      | ✓                    |                      |
| Year FEs                               | ✓                    | ✓                    |                      |
| Province $\times$ Year FEs             |                      |                      | ✓                    |

*Notes:* This table reports heterogeneity in the COVID-19 effect by gender. The dependent variable is an indicator for out-of-hometown migrant work. The base category is male workers, so the coefficient on Non-Agri  $\times$  COVID-19 reflects the effect for men. The interaction term captures the differential effect for women. Column (1) includes individual and year fixed effects; Column (2) includes household and year fixed effects; Column (3) includes individual and province-by-year fixed effects. Standard errors clustered at the individual level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.13: Heterogeneity: The Effect of COVID-19 by Age Group

| VARIABLES                                   | Out-of-Hometown Work |                      |                      |
|---------------------------------------------|----------------------|----------------------|----------------------|
|                                             | (1)                  | (2)                  | (3)                  |
| <i>Base Effect (Age 35–49)</i>              |                      |                      |                      |
| Non-Agri $\times$ COVID-19                  | -0.082***<br>(0.013) | -0.084***<br>(0.014) | -0.096***<br>(0.013) |
| <i>Interactions (Relative to Age 35–49)</i> |                      |                      |                      |
| $\times$ Age 16–34                          | 0.036<br>(0.024)     | 0.035<br>(0.027)     | 0.039<br>(0.024)     |
| $\times$ Age 50–60                          | -0.040**<br>(0.019)  | -0.035*<br>(0.021)   | -0.043**<br>(0.018)  |
| Observations                                | 33,152               | 32,666               | 33,014               |
| R-squared                                   | 0.630                | 0.478                | 0.636                |
| Individual FEs                              | ✓                    |                      | ✓                    |
| Household FEs                               |                      | ✓                    |                      |
| Year FEs                                    | ✓                    | ✓                    |                      |
| Province $\times$ Year FEs                  |                      |                      | ✓                    |

*Notes:* This table reports heterogeneity in the COVID-19 effect by age group. The dependent variable is an indicator for out-of-hometown migrant work. The base category is prime-age workers (ages 35–49), so the coefficient on Non-Agri  $\times$  COVID-19 reflects the effect for this group. Interaction terms capture the differential effects for younger workers (ages 16–34) and older workers (ages 50–60). Column (1) includes individual and year fixed effects; Column (2) includes household and year fixed effects; Column (3) includes individual and province-by-year fixed effects. Standard errors clustered at the individual level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.14: Robustness: Using Prefecture-Level COVID-19 Cases

| VARIABLES                             | Out-of-Hometown Work |                      |                      |
|---------------------------------------|----------------------|----------------------|----------------------|
|                                       | (1)                  | (2)                  | (3)                  |
| Non-Agri $\times$ ln(COVID cases + 1) | -0.013***<br>(0.002) | -0.019***<br>(0.002) | -0.018***<br>(0.002) |
| Observations                          | 33,986               | 33,510               | 33,856               |
| R-squared                             | 0.631                | 0.454                | 0.636                |
| Individual FEs                        | ✓                    |                      | ✓                    |
| Household FEs                         |                      | ✓                    |                      |
| Year FEs                              | ✓                    | ✓                    |                      |
| Province $\times$ Year FEs            |                      |                      | ✓                    |

*Notes:* This table reports robustness checks using a continuous measure of COVID-19 intensity. The dependent variable is an indicator for out-of-hometown migrant work. The shock variable is the natural logarithm of one plus the number of confirmed COVID-19 cases at the prefecture-year level. Column (1) includes individual and year fixed effects; Column (2) includes household and year fixed effects; Column (3) includes individual and province-by-year fixed effects. Standard errors clustered at the individual level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.15: Effect of COVID-19 Cases on Automation Innovation

| VARIABLES                              | Robot<br>Patent %<br>(1) | Robot<br>Count<br>(2) | ln(Robot<br>Count+1)<br>(3) | Robot per<br>Capita<br>(4) | IHS(Robot<br>Count)<br>(5) | ln(Total<br>Patent+1)<br>(6) | ln(Non-Auto<br>Patent+1)<br>(7) |
|----------------------------------------|--------------------------|-----------------------|-----------------------------|----------------------------|----------------------------|------------------------------|---------------------------------|
| Migrant Share %<br>× ln(COVID Cases+1) | 0.009**<br>(0.004)       | 2.059***<br>(0.418)   | 0.008***<br>(0.003)         | 0.028***<br>(0.005)        | 0.007**<br>(0.003)         | 0.002<br>(0.002)             | 0.002<br>(0.002)                |
| Observations                           | 21,568                   | 21,568                | 21,568                      | 21,568                     | 21,568                     | 21,568                       | 21,568                          |
| R-squared                              | 0.219                    | 0.830                 | 0.863                       | 0.826                      | 0.850                      | 0.943                        | 0.942                           |
| County FEs                             | ✓                        | ✓                     | ✓                           | ✓                          | ✓                          | ✓                            | ✓                               |
| Province × Year FEs                    | ✓                        | ✓                     | ✓                           | ✓                          | ✓                          | ✓                            | ✓                               |
| Initial GDP × Year FEs                 | ✓                        | ✓                     | ✓                           | ✓                          | ✓                          | ✓                            | ✓                               |

*Notes:* This table reports heterogeneous effects of COVID-19 intensity on automation patent innovation by county-level migrant share. The interaction term captures how counties with higher pre-period migrant shares respond differentially to COVID-19 exposure (measured as the natural logarithm of one plus confirmed 2020 prefecture-level cases). Columns 1–5 report outcomes related to robot/automation patents; Columns 6–7 report broader patent counts as placebo checks. All specifications include county fixed effects, province × year fixed effects, and interactions between initial GDP per capita and year fixed effects. Standard errors clustered at the county level in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.10.

Table C.16: COVID-19 Effect on Robot Installations: China vs. Rest of World

| VARIABLES                                   | Overall<br>(1)      | Manufacturing/<br>Construction<br>(2) | Automotive<br>(3)   | Labor-<br>intensive<br>(4) | High-tech<br>(5)     | Electronics &<br>Electrical<br>(6) |
|---------------------------------------------|---------------------|---------------------------------------|---------------------|----------------------------|----------------------|------------------------------------|
| <i>Panel A: China's Trend</i>               |                     |                                       |                     |                            |                      |                                    |
| China × COVID-19                            | 1.201***<br>(0.039) | 0.558***<br>(0.050)                   | 1.025***<br>(0.042) | 1.174***<br>(0.039)        | 1.287***<br>(0.036)  | 1.239***<br>(0.037)                |
| <i>Panel B: Group Specific Coefficients</i> |                     |                                       |                     |                            |                      |                                    |
| × Group                                     |                     | 0.713***<br>(0.034)                   | 0.781***<br>(0.031) | 0.213***<br>(0.030)        | −0.445***<br>(0.033) | −0.198***<br>(0.041)               |
| Observations                                | 27,900              | 27,900                                | 27,900              | 27,900                     | 27,900               | 27,900                             |
| R-squared                                   | 0.724               | 0.724                                 | 0.725               | 0.724                      | 0.725                | 0.725                              |
| Country FEs                                 | ✓                   | ✓                                     | ✓                   | ✓                          | ✓                    | ✓                                  |
| Industry FEs                                | ✓                   | ✓                                     | ✓                   | ✓                          | ✓                    | ✓                                  |
| Year FEs                                    | ✓                   | ✓                                     | ✓                   | ✓                          | ✓                    | ✓                                  |

*Notes:* This table reports difference-in-differences estimates for robot installations comparing China to the rest of the world. Panel A reports the baseline China × COVID-19 coefficient from each regression. Panel B reports the triple-interaction term (China × COVID-19 × Group), capturing the additional effect in each sector group relative to the remaining industries. Column (1) reports the overall effect without a group interaction. Columns (2)–(6) correspond to group-specific regressions; lower-order interactions are included but not reported. The groups aggregate the detailed IFR industries as follows: Manufacturing/Construction comprises all industries except agriculture, utilities, and education/R&D; Automotive comprises vehicles and all automotive parts; Labor-intensive comprises food and beverages, textiles, wood and furniture, and paper; High-tech comprises semiconductors, electronic components, computers, information and communication equipment, medical instruments, and pharmaceuticals; Electronics & Electrical comprises semiconductors, electronic components, computers, information and communication equipment, electrical machinery, and household appliances. The High-tech and Electronics & Electrical groups overlap in four industries, so columns (5) and (6) are not independent tests. All specifications include country, industry, and year fixed effects. Robot installation data are from the International Federation of Robotics (IFR); the estimation sample is restricted to the baseline analysis period, 2015–2023. Standard errors are clustered by country in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.10.

Table C.17: COVID-19 Effect on Robot Stocks: China vs. Rest of World

| VARIABLES                                   | Overall<br>(1)      | Manufacturing/<br>Construction<br>(2) | Automotive<br>(3)    | Labor-<br>intensive<br>(4) | High-tech<br>(5)    | Electronics &<br>Electrical<br>(6) |
|---------------------------------------------|---------------------|---------------------------------------|----------------------|----------------------------|---------------------|------------------------------------|
| <i>Panel A: China's Trend</i>               |                     |                                       |                      |                            |                     |                                    |
| China $\times$ COVID-19                     | 0.960***<br>(0.032) | 1.129***<br>(0.038)                   | 1.046***<br>(0.035)  | 0.920***<br>(0.034)        | 0.976***<br>(0.029) | 0.973***<br>(0.028)                |
| <i>Panel B: Group Specific Coefficients</i> |                     |                                       |                      |                            |                     |                                    |
| $\times$ Group                              |                     | -0.187***<br>(0.042)                  | -0.385***<br>(0.029) | 0.306***<br>(0.036)        | -0.086**<br>(0.041) | -0.072<br>(0.047)                  |
| Observations                                | 27,900              | 27,900                                | 27,900               | 27,900                     | 27,900              | 27,900                             |
| R-squared                                   | 0.829               | 0.829                                 | 0.829                | 0.829                      | 0.830               | 0.830                              |
| Country FEs                                 | ✓                   | ✓                                     | ✓                    | ✓                          | ✓                   | ✓                                  |
| Industry FEs                                | ✓                   | ✓                                     | ✓                    | ✓                          | ✓                   | ✓                                  |
| Year FEs                                    | ✓                   | ✓                                     | ✓                    | ✓                          | ✓                   | ✓                                  |

*Notes:* This table reports difference-in-differences estimates for robot stocks comparing China to the rest of the world. Panel A reports the baseline China  $\times$  COVID-19 coefficient from each regression. Panel B reports the triple-interaction term (China  $\times$  COVID-19  $\times$  Group), capturing the additional effect in each sector group relative to the remaining industries. Column (1) reports the overall effect without a group interaction. Columns (2)–(6) correspond to group-specific regressions; lower-order interactions are included but not reported. The groups aggregate the detailed IFR industries as follows: Manufacturing/Construction comprises all industries except agriculture, utilities, and education/R&D; Automotive comprises vehicles and all automotive parts; Labor-intensive comprises food and beverages, textiles, wood and furniture, and paper; High-tech comprises semiconductors, electronic components, computers, information and communication equipment, medical instruments, and pharmaceuticals; Electronics & Electrical comprises semiconductors, electronic components, computers, information and communication equipment, electrical machinery, and household appliances. The High-tech and Electronics & Electrical groups overlap in four industries, so columns (5) and (6) are not independent tests. All specifications include country, industry, and year fixed effects. Robot stock data are from the International Federation of Robotics (IFR); the estimation sample is restricted to the baseline analysis period, 2015–2023. Standard errors are clustered by country in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.18: Robustness: China's Post-COVID Robot Installations Against Alternative Control Groups

|                          | (1)                    | (2)                 | (3)                 | (4)                | (5)                  |
|--------------------------|------------------------|---------------------|---------------------|--------------------|----------------------|
|                          | All other<br>countries | Emerging<br>Asia    | Emerging<br>global  | East Asia<br>hubs  | Pre-trend<br>matched |
| China $\times$ COVID-19  | 1.201***<br>(0.039)    | 0.610***<br>(0.148) | 0.705***<br>(0.112) | 0.685**<br>(0.326) | 0.716***<br>(0.078)  |
| Observations             | 27,900                 | 1,674               | 2,511               | 1,395              | 3,069                |
| R-squared                | 0.724                  | 0.763               | 0.683               | 0.686              | 0.658                |
| No. of control countries | 99                     | 5                   | 8                   | 4                  | 10                   |
| Country FEs              | ✓                      | ✓                   | ✓                   | ✓                  | ✓                    |
| Industry FEs             | ✓                      | ✓                   | ✓                   | ✓                  | ✓                    |
| Year FEs                 | ✓                      | ✓                   | ✓                   | ✓                  | ✓                    |

*Notes:* This table re-estimates the aggregate China  $\times$  COVID-19 difference-in-differences for log robot installations, varying the control group. Column (1) uses the full IFR sample, identical to the baseline specification in Table C.16; the country count includes the IFR regional reporting units. Column (2) restricts the control group to emerging manufacturing Asia: Vietnam, India, Thailand, Malaysia, and Indonesia. Column (3) adds the major emerging manufacturing exporters outside Asia: Mexico, Brazil, and Türkiye. Column (4) uses the East Asian manufacturing hubs: Japan, the Republic of Korea, Chinese Taipei, and Singapore. Column (5) uses a data-driven control group: the ten economies whose pre-pandemic installation growth (2015 to 2019) is closest to China's, namely Belgium, Brazil, France, Hungary, India, Italy, Poland, Singapore, Switzerland, and Vietnam. Columns (2)–(5) rest on 4–10 country clusters, so their standard errors should be interpreted with caution. All specifications include country, industry, and year fixed effects; IFR data, 2015–2023. Standard errors clustered by country in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .



Table C.19: Robustness: China's Post-COVID Robot Stocks Against Alternative Control Groups

|                          | (1)                 | (2)                | (3)                 | (4)                 | (5)                 |
|--------------------------|---------------------|--------------------|---------------------|---------------------|---------------------|
|                          | All other countries | Emerging Asia      | Emerging global     | East Asia hubs      | Pre-trend matched   |
| China $\times$ COVID-19  | 0.960***<br>(0.032) | 0.260**<br>(0.124) | 0.402***<br>(0.106) | 0.757***<br>(0.144) | 0.718***<br>(0.134) |
| Observations             | 27,900              | 1,674              | 2,511               | 1,395               | 3,069               |
| R-squared                | 0.829               | 0.844              | 0.800               | 0.705               | 0.759               |
| No. of control countries | 99                  | 5                  | 8                   | 4                   | 10                  |
| Country FEs              | ✓                   | ✓                  | ✓                   | ✓                   | ✓                   |
| Industry FEs             | ✓                   | ✓                  | ✓                   | ✓                   | ✓                   |
| Year FEs                 | ✓                   | ✓                  | ✓                   | ✓                   | ✓                   |

*Notes:* This table re-estimates the aggregate China  $\times$  COVID-19 difference-in-differences for log operational robot stocks, varying the control group. Column (1) uses the full IFR sample, identical to the baseline specification in Table C.17; the country count includes the IFR regional reporting units. Column (2) restricts the control group to emerging manufacturing Asia: Vietnam, India, Thailand, Malaysia, and Indonesia. Column (3) adds the major emerging manufacturing exporters outside Asia: Mexico, Brazil, and Türkiye. Column (4) uses the East Asian manufacturing hubs: Japan, the Republic of Korea, Chinese Taipei, and Singapore. Column (5) uses a data-driven control group: the ten economies whose pre-pandemic installation growth (2015 to 2019) is closest to China's, namely Belgium, Brazil, France, Hungary, India, Italy, Poland, Singapore, Switzerland, and Vietnam. Columns (2)–(5) rest on 4–10 country clusters, so their standard errors should be interpreted with caution. All specifications include country, industry, and year fixed effects; IFR data, 2015–2023. Standard errors clustered by country in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.20: More Robustness Checks

| VARIABLES                                                                                        | Robot Patent<br>Share %<br>(1) | Robot<br>Count<br>(2) | ln(Robot<br>Count+1)<br>(3) | Robot per<br>Capita<br>(4) | IHS(Robot<br>Count)<br>(5) |
|--------------------------------------------------------------------------------------------------|--------------------------------|-----------------------|-----------------------------|----------------------------|----------------------------|
| <i>Panel A: Weighted by Population Size</i>                                                      |                                |                       |                             |                            |                            |
| Migrant Share % $\times$ COVID-19                                                                | 0.039**<br>(0.019)             | 11.676***<br>(2.819)  | 0.033***<br>(0.013)         | 0.143***<br>(0.028)        | 0.027*<br>(0.015)          |
| Observations                                                                                     | 21,568                         | 21,568                | 21,568                      | 21,568                     | 21,568                     |
| R-squared                                                                                        | 0.255                          | 0.836                 | 0.879                       | 0.835                      | 0.865                      |
| <i>Panel B: Excluding Provincial Capital Cities</i>                                              |                                |                       |                             |                            |                            |
| Migrant Share % $\times$ COVID-19                                                                | 0.051**<br>(0.021)             | 11.374***<br>(3.587)  | 0.035**<br>(0.014)          | 0.136***<br>(0.035)        | 0.032*<br>(0.016)          |
| Observations                                                                                     | 19,166                         | 19,166                | 19,166                      | 19,166                     | 19,166                     |
| R-squared                                                                                        | 0.235                          | 0.838                 | 0.837                       | 0.842                      | 0.821                      |
| <i>Panel C: Excluding Deputy Provincial-Level Cities</i>                                         |                                |                       |                             |                            |                            |
| Migrant Share % $\times$ COVID-19                                                                | 0.040**<br>(0.020)             | 10.763***<br>(3.248)  | 0.032**<br>(0.014)          | 0.128***<br>(0.032)        | 0.027*<br>(0.016)          |
| Observations                                                                                     | 20,632                         | 20,632                | 20,632                      | 20,632                     | 20,632                     |
| R-squared                                                                                        | 0.247                          | 0.848                 | 0.864                       | 0.843                      | 0.849                      |
| <i>Panel D: Excluding Provincial Capitals and Deputy Provincial-Level Cities</i>                 |                                |                       |                             |                            |                            |
| Migrant Share % $\times$ COVID-19                                                                | 0.047**<br>(0.023)             | 11.425***<br>(3.988)  | 0.030**<br>(0.014)          | 0.133***<br>(0.038)        | 0.027<br>(0.017)           |
| Observations                                                                                     | 18,935                         | 18,935                | 18,935                      | 18,935                     | 18,935                     |
| R-squared                                                                                        | 0.234                          | 0.836                 | 0.826                       | 0.841                      | 0.810                      |
| <i>Panel E: Alternative Clustering at Province-Year Level</i>                                    |                                |                       |                             |                            |                            |
| Migrant Share % $\times$ COVID-19                                                                | 0.063***<br>(0.018)            | 5.457***<br>(1.309)   | 0.035***<br>(0.009)         | 0.081***<br>(0.018)        | 0.034***<br>(0.010)        |
| Observations                                                                                     | 21,568                         | 21,568                | 21,568                      | 21,568                     | 21,568                     |
| R-squared                                                                                        | 0.219                          | 0.826                 | 0.863                       | 0.822                      | 0.850                      |
| <i>Panel F: Controlling for County Non-Agricultural Value Added <math>\times</math> Year FEs</i> |                                |                       |                             |                            |                            |
| Migrant Share % $\times$ COVID-19                                                                | 0.056***<br>(0.021)            | 5.578***<br>(1.384)   | 0.025**<br>(0.011)          | 0.078***<br>(0.016)        | 0.023*<br>(0.013)          |
| Observations                                                                                     | 21,242                         | 21,242                | 21,242                      | 21,242                     | 21,242                     |
| R-squared                                                                                        | 0.218                          | 0.829                 | 0.861                       | 0.829                      | 0.847                      |
| County FEs                                                                                       | ✓                              | ✓                     | ✓                           | ✓                          | ✓                          |
| Province $\times$ Year FEs                                                                       | ✓                              | ✓                     | ✓                           | ✓                          | ✓                          |
| Initial GDP $\times$ Year FEs                                                                    | ✓                              | ✓                     | ✓                           | ✓                          | ✓                          |

*Notes:* This table presents robustness checks for the baseline estimates of the effect of labor mobility restrictions on automation patent innovation, interacted with county-level migrant share. COVID-19 is a binary indicator equal to one in the post-onset period. The interaction term captures how counties with higher pre-period migrant shares differentially increase automation adoption following COVID-19 onset. All panels use the fully saturated specification (county FEs, province  $\times$  year FEs, and initial economic conditions  $\times$  year FEs). Panel A reweights observations by county population size. Panel B excludes counties belonging to provincial capital cities. Panel C excludes counties belonging to deputy provincial-level cities. Panel D applies both exclusions simultaneously. Panel E re-estimates the baseline using standard errors clustered at the province-year level rather than the county level. Panel F adds interactions between county-level non-agricultural value added (2015) and year fixed effects to control for differential trends in industrial composition. Standard errors clustered at the county level in parentheses (except Panel E). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table C.21: [Diegert et al. \(2022\)](#) Sensitivity Analysis: R&D-Channel Calibrations

|                                                          | Anchor<br>$W_1 = \text{patents}$ | Robustness 1<br>$W_1 = \text{patents} + \text{S\&T}$ | Robustness 2<br>$W_1 = \text{patents} + \text{firm R\&D}$ |
|----------------------------------------------------------|----------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| $\hat{\beta}$ (full)                                     | 0.0704<br>(0.0199)               | 0.0529<br>(0.0204)                                   | 0.0452<br>(0.0174)                                        |
| $\hat{\beta}$ (short, no $W_1$ )                         | 0.0701                           | 0.0525                                               | 0.0449                                                    |
| $N$                                                      | 21,568                           | 16,514                                               | 15,196                                                    |
| <i>Breakdown point (<math>\widehat{rxbar}</math>, %)</i> |                                  |                                                      |                                                           |
| $\bar{c} = 1$                                            | 89.7                             | 36.1                                                 | 90.2                                                      |
| $\bar{c} = 0.5$                                          | 107.9                            | 36.1                                                 | 109.4                                                     |
| $\bar{c} = 0$                                            | 203.1                            | 38.7                                                 | 209.1                                                     |

Notes: This table reports [Diegert et al. \(2022\)](#) sensitivity analysis results for the coefficient on migrant exposure interacted with the post-COVID-19 indicator. All specifications partial out county and province-by-year fixed effects ( $W_0$ ). Calibration covariates ( $W_1$ ) are residualized on  $W_0$  and differ by column: the anchor specification uses county log patent stock; Robustness 1 adds prefecture science and technology expenditure; Robustness 2 adds listed-firm R&D spending.  $\hat{\beta}$  (full) is the coefficient with  $W_1$  included;  $\hat{\beta}$  (short) is the coefficient with  $W_1$  excluded, on the same estimation sample. Standard errors clustered at the county level in parentheses. The breakdown point is the value of  $rxbar$  at which the identified set first includes zero, reported under three assumptions on  $\bar{c}$ , the maximum correlation between the unobserved confounder and  $W_1$ .

Table C.22: Effect of Migration Restrictions on Labor Market Reallocation and Occupational Adjustment

| VARIABLES                                      | Informal Employment |                     | Unemployment     |                    | ln(Income)         |                    | Switch from Mfg   |                   |
|------------------------------------------------|---------------------|---------------------|------------------|--------------------|--------------------|--------------------|-------------------|-------------------|
|                                                | (1)                 | (2)                 | (3)              | (4)                | (5)                | (6)                | (7)               | (8)               |
| Migrant Worker <sub>pre-COVID</sub> × COVID-19 | 0.043***<br>(0.009) | 0.041***<br>(0.009) | 0.010<br>(0.006) | 0.013**<br>(0.006) | -0.079*<br>(0.045) | -0.090*<br>(0.049) | -0.010<br>(0.013) | -0.012<br>(0.013) |
| Observations                                   | 38,298              | 38,298              | 38,298           | 38,298             | 16,878             | 16,876             | 13,770            | 13,766            |
| R-squared                                      | 0.843               | 0.844               | 0.578            | 0.581              | 0.604              | 0.608              | 0.384             | 0.389             |
| Individual FEs                                 | ✓                   | ✓                   | ✓                | ✓                  | ✓                  | ✓                  | ✓                 | ✓                 |
| Year FEs                                       | ✓                   |                     | ✓                |                    | ✓                  |                    | ✓                 |                   |
| Province × Year FEs                            |                     | ✓                   |                  | ✓                  |                    | ✓                  |                   | ✓                 |

*Notes:* This table reports the effect of COVID-19-induced migration restrictions on labor market reallocation and occupational adjustment. The key independent variable is the interaction between pre-pandemic migrant worker status and COVID-19 onset. Columns (1) and (2) report results for informal employment; columns (3) and (4) for unemployment; columns (5) and (6) for migrant worker's income; and columns (7) and (8) for switching out of manufacturing. All specifications include individual fixed effects. Odd-numbered columns include year fixed effects; even-numbered columns replace year fixed effects with Province × Year fixed effects. Standard errors clustered at the county level in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.10.

Table C.23: Back-of-the-Envelope Aggregate Income Losses of the Automation Response

| Channel                          | Affected workers  | Annual loss (bn yuan) | % of GDP     |
|----------------------------------|-------------------|-----------------------|--------------|
| Migrant income loss (8.5%)       | 170 million       | 751                   | 0.74%        |
| Informality wage penalty (17.5%) | 7 million         | 64                    | 0.06%        |
| <b>Total labor market cost</b>   |                   | <b>815</b>            | <b>0.80%</b> |
| <i>Memo: Capital deepening</i>   | <i>12 million</i> | <i>141</i>            | <i>0.14%</i> |

*Notes:* Back-of-the-envelope estimates combining regression coefficients from Tables C.22 and C.6 with aggregate population and earnings statistics. The income loss row applies the midpoint estimate of 8.5 percent (from columns 5 and 6 of Table C.22) to 170 million migrant workers in treatment counties earning an average of 52,000 yuan annually (NBS Migrant Workers Monitoring Survey, 2020). The informality row applies a 17.5 percent informal-sector wage penalty to the 4.2 percentage point increase in informality probability, implying approximately 7 million newly informal workers. The capital deepening row scales the fixed-asset-per-employee estimate from Table C.6 to the 12 million employees covered by the CSMAR listed-firm sample and is reported as a memo item because it reflects firm-level investment rather than a net social loss; whether it offsets the worker-side income losses depends on the long-run productivity returns to automation. The GDP denominator is China's 2020 nominal GDP of 101.6 trillion yuan (NBS). All figures are approximate and should be interpreted as order-of-magnitude benchmarks.